

**Design and Implementation of an Interoperable Risk based Food
Safety Inspection Information Management System (IIMS) for the
FCAU of Ministry of Health**

SOFTWARE REQUIREMENTS SPECIFICATION

Prepared By



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Key Summary

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Introduction

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the design and implementation of the Interoperable Risk-Based Food Safety Inspection Information Management System (IIMS) for the Food Control Administration Unit (FCAU) of the Ministry of Health. The system is intended to serve as a centralized, web-based platform that supports the end-to-end lifecycle of food safety regulation, including premises registration, inspection and grading, laboratory coordination, enforcement actions, and import/export control processes.

The SRS provides a comprehensive description of system behavior, workflows, data structures, and interactions among various stakeholders such as Food Business Operators (FBOs), Public Health Inspectors (PHIs), Medical Officers of Health (MOHs), laboratory personnel, and administrative authorities. It establishes a common understanding between stakeholders, developers, and project sponsors, ensuring that the system is developed in alignment with regulatory requirements, operational needs, and technological standards.

Document Purpose

The purpose of this document is to formally capture and communicate the complete set of requirements for the IIMS, serving as a foundational reference throughout the system development lifecycle. It outlines the system scope, detailed functional processes, data models, and non-functional requirements such as performance, security, interoperability, availability, usability, and maintainability.

This document is intended to:

- Provide a clear and structured specification for system design and development
- Serve as a baseline for validation, testing, and acceptance of the system
- Facilitate communication among stakeholders, including technical teams, domain experts, and governing authorities
- Ensure traceability of requirements from business needs to implemented features
- Support future system enhancements and maintenance activities

By defining these requirements in a structured and standardized manner, the SRS ensures that the resulting system is robust, scalable, and aligned with the national food safety regulatory framework.

1. System Overview

The Food Control Administration Unit (FCAU) requires a comprehensive, modern, and fully integrated **Web Based Interoperable Risk based Food Safety Inspection Information Management System** to strengthen and streamline food safety governance across Sri Lanka. The proposed system will serve as a centralized digital platform to manage the end-to-end lifecycle of food premises regulation, inspection, sampling, laboratory coordination, enforcement actions, food import control, and regulatory reporting. Designed with scalability, transparency, and accountability in mind, the system will replace fragmented and manual processes with a structured, role based, and workflow driven solution aligned with national food safety regulations and administrative structures.

The system will consist of the following core modules:

- User and Access Management Module
- Premises Registration and Lifecycle Management Module
- Inspection and Grading Module
- Laboratory Coordination Module
- Legal, Prosecution and Enforcement Management Module
- Import Control Module
- Export Control Module
- Registration of Common Salt Manufacturing and Transports Management Module
- Bottled Water Manufacturing and Registration Management Module
- Risk based Food Safety Inspection via Risk Analysis Framework
- Complaint Management Module
- Alerts and Notifications Module
- Reports Module

2. User and Access Management Module

The User Management Module (UMM) serves as the foundational security, authentication, and identity management layer for the Food Safety Inspection Information Management System. The UMM handles user registration, authentication, authorization, and profile management across the entire digital ecosystem. The module ensures secure, interoperable, and centralized access for all regulatory actors.

2.1 Authentication and Identity Management

To streamline access and enhance security across web and mobile platforms, the UMM implements secured Sign in;

- **MFA Integration:** The system balances user convenience and enterprise grade security by combining with Multi-Factor Authentication (MFA)
- **Credential Policies:** The system enforces mandatory credential updates upon first login and requires periodic authentication renewals.

2.2 Role-Based Access Control (RBAC) Architecture

The system utilizes a strictly defined Role-Based Access Control (RBAC) model to restrict system access based on predefined job functions rather than individual user identities. This simplifies access management and significantly reduces the risk of unauthorized data exposure.

- **Principle of Least Privilege:** Users are assigned the fewest number of permissions necessary to execute their regulatory duties.
- **Primary Role Definitions:**
 - *FCAU Administrator:* Granted global system configuration rights, risk matrix adjustment capabilities, and access to macro-level national compliance dashboards.
 - *Medical Officer of Health (MOH):* Granted regional oversight, allowing them to assign inspection tasks, lock non-compliant establishments, and review localized analytical data.
 - *Public Health Inspector (PHI):* Granted field-level access to for digital H-800 form submission, establishment profiling, and sample registry entry strictly within their designated geographic division.

- *Laboratory Analysts:* Granted restricted API access to upload analytical test results to the sample custody register.
- And for all other users, depending on the discussed work flows below, system access would be granted and that is configurable.

2.2.1 Login Security

- **Access and Audit Controls:** The system implements stringent software and procedural mechanisms to record and examine all access events and activities involving sensitive operational or personnel data.
- **Encryption:** All sensitive user data, credentials, and session tokens must be encrypted in transit using two-way SSL/TLS authentication and encrypted at rest within the database.
- **Integrity Checks:** E-signatures and integrity checks are enforced to prevent data tampering or hostile object creation by compromised accounts.

2.3 Identity Inception and Progressive Profiling

The initial interaction a user undertakes within the system occurs during the registration and onboarding phase. The system must provide a robust, friction free self registration framework that enables users to establish their digital identity while capturing essential authentication metadata. Because modern applications frequently suffer from high user abandonment rates during lengthy registration forms, the module must implement a multi-level, progressive onboarding process that stages data collection.

In the primary stage (Level 1), the system mandates the collection of core identity details strictly necessary for account creation, such as the user's first name, last name, primary email address, mobile number, and initial cryptographic credentials. Following the successful submission of this baseline data, the system immediately initiates account activation protocols. This workflow utilizes out-of-band verification, dispatching a cryptographic challenge via an email verification link to validate the ownership of the provided communication channels. Once baseline identity is authenticated, the system transitions the user into secondary and tertiary onboarding levels.

2.4 Administrative Oversight and Platform Governance

While individual entities manage their localized data, authorized system administrators require a panoptic view of the platform's landscape. The administrative functional partition must provide a comprehensive user management interface dedicated to overseeing all core entities. Administrators must be equipped with granular capabilities to view, manually approve, temporarily ban, or permanently deactivate user accounts based on behavioral policy violations or security anomalies.

Administrative oversight heavily involves systemic governance and credential lifecycle management. Support personnel must have the functional capability to securely trigger password resets, revoke active sessions, and monitor the real-time status of user accounts to provide responsive technical assistance.

3. Premises Registration and Life Cycle Management Module

The Premises Registration and Lifecycle Management Module encompasses the complete lifecycle of a premises registration process, beginning from user onboarding and application submission to inspection, approval, rejection, and appeal handling. The system supports classification of premises based on regulatory categories, risk levels, and registration requirements as defined by FCAU policies.

In addition to handling new registrations, the module is designed to maintain a persistent record of premises throughout their lifecycle, enabling updates, re-evaluations, and compliance tracking. It integrates multiple stakeholders into a unified workflow, ensuring that each decision point is governed by predefined rules and that all transitions between stages are properly controlled and documented.

3.1 Definitions and Terminology

Within the context of this system, several domain-specific terms are used. A “Premises” refers to any physical location engaged in food-related activities requiring regulatory oversight. The “GN Division” denotes the Grama Niladhari administrative division used for geographical mapping and officer assignment. “Risk Categorization” refers to the classification of premises based on potential food safety risks. “Inspection” denotes the physical verification process conducted by PHIs, while “Appeal” represents a formal request by an applicant to review a rejected decision.

3.2 Workflow Elaboration and State Machine Architecture

The architectural flow of the Premises Registration module is governed by a precise sequence of human-computer interactions and automated background processes. The workflow diagram dictates, illustrating the pathways of five primary actors:

- Food Business Operator (FBO),
- Public Health Inspector (PHI),
- Medical Officer of Health (MOH),
- Appeal Board, and
- Regional Director (RD).

The orchestration of these actors is supported by five distinct computational engines that execute in the background.

Phase 01: Front-End Onboarding and Profile Initialization

The lifecycle commences with the FBO interacting with the citizen-facing portal. The operator is required to "Sign up to the system," establishing authenticated credentials. Upon successful authentication, the system executes the "Create Temporary Profile" subroutine. This architectural distinction is critical: a temporary profile represents an unverified entity, shielding the core authoritative database from unvalidated or fraudulent entries.

The operator proceeds to "Login to the Profile" and initiates the "Fill the Registration Form" process. At this juncture, the application exists in a DRAFT state, allowing the user to incrementally upload the mandatory statutory documents required by the Regulations, such as business registration certificates, and other required attributes. Once the data matrix is complete, the operator triggers the "Submit the Form" action, transitioning the state to SUBMITTED.

Phase 02: Automated Background Engines and Spatial Routing

Simultaneous to the submission, the Laravel backend queues the payload through a sequence of five specialized validation and categorization engines. This automated orchestration minimizes manual administrative overhead and prevents systemic data corruption.

- **GN Mapping Engine:** Sri Lanka's administrative geography is divided into Grama Niladhari (GN) divisions. The GN Mapping engine parses the physical address provided by the applicant and links the application to a specific spatial polygon. This is the cornerstone of the system's routing logic, ensuring that the information is exclusively visible to the MOH and PHI whose jurisdiction covers that specific GN division.
- **Duplicate Validation Engine:** Before proceeding, the payload is parsed against existing records to prevent regulatory overlap. Utilizing algorithms to cross-reference the National Identity Card (NIC) of the owner, the Business Registration Number, and the exact spatial coordinates, the engine flags potential duplicates.
- **Premises Types Engine:** The system evaluates the nature of the business to categorize it into one of three primary regulatory streams: "Registered," "Registration Not Required," or "Unregistered."

- **Premises Categorization - Regulation:** For premises requiring registration, the engine assigns the entity to one of the 10 statutory categories defined in the regulations (e.g., Factory, Bakery, Hotel/Restaurant, Snack Bar, Supermarket).
- **Premises Categorization - Risk-based:** The engine applies an algorithmic assessment to determine the epidemiological risk profile of the establishment, dictating the urgency and frequency of subsequent inspections.

Phase 03: Primary Review and Multi-Tiered Inspection Logic

Once the automated engines yield a validated SUBMITTED state, the information enters the operational queue of the MOH. The MOH performs the "Get the Application based on GN division assigned" action and executes a preliminary review before invoking the "Send to PHI" action. The system concurrently allows for an alternate entry point: a PHI may "Login to the System" and "Initiate registration on behalf of the premises" for applicants lacking digital literacy, advancing the application directly to the review stage.

The PHI receives the information via the "Get the Application based on GN division assigned" interface. The state transitions to REVIEW_STATUS. Upon documentary satisfaction, the PHI conducts the "1st Visit." This physical inspection is quantified using the standardized Form H-800 grading checklist. The outcome of the 1st Visit leads to a critical decision node.

- **Approval Path:** If the PHI determines the premises is compliant, the decision branch flows to "Approve." The system routes the application back to the MOH, who executes "Approve the application" and finalizes the process by triggering the "Create the premises profile" module, shifting the temporary entity to a permanent, legally recognized standing.
- **Rejection and 2nd Visit Path:** If the 1st Visit reveals non-compliance, the PHI selects "Reject." The application state shifts to PENDING_2ND_VISIT. The FBO is granted a remediation window, followed by a mandatory "2nd Visit" by the PHI. If the 2nd Visit yields an "Approve" decision, the flow mirrors the primary approval path to the MOH. If the 2nd Visit yields another "Reject," the system terminates the primary inspection loop and automatically triggers a notification node, officially communicating the rejection to the applicant.

Phase 04: The Grievance and Appeal Workflow

A rejected applicant possesses the legal right to invoke the "Appeal" node. Activating this workflow shifts the application state to APPEAL_PENDING. The system routes the information to the "Appeal Board" interface, triggering the "Assign Appeal Board" subroutine. The board conducts an independent review and arrives at an "Approve / Reject" decision node.

Regardless of the Appeal Board's preliminary finding, the workflow demands that the information is "Sent to RD" (Regional Director). The RD "receives Appeal" and executes a "Review and Approve" function, leading to the "Final Approval" diamond.

- **Final Approval - Reject:** The RD rejects the appeal. The system routes this to a node where "MOH receives RD's Decision." The MOH acknowledges the rejection, and the system issues a terminal failure notification to the applicant.
- **Final Approval - Approve:** The RD overturns the lower-level rejection. The information flows to the "MOH receives RD's Decision" node. The MOH is structurally compelled to accept the appellate ruling, executing "Approve the application" and invoking the "Create the premises profile" subroutine. Simultaneously, the system ensures the "Applicant receives RD's Decision Approved or Reject same time both receive notifications" node is satisfied, ensuring absolute transparency.

3.3 Statutory Inspection Standards: Form H-800 Integration

The core mechanism for verifying compliance during the 1st and 2nd visits is the digitization of the Food Handling Trades Inspection Rating Form, legally recognized as the H-800. The Laravel application must programmatically enforce the parameters of this grading checklist to automate the calculation of the final risk score.

The system will feature a mobile-responsive interface for the PHI to input evaluation metrics across ten statutorily defined categories, frequently referred to as "cages":

1. **Location and Environment:** Evaluation of external vectors, including freedom from industrial pollution, objectionable odors, and insect or animal infestations.
2. **Building Infrastructure:** Assessment of sound structural construction, the provision of adequate working space, lighting, and mechanical ventilation.

3. **Processing Room Dynamics:** Measurement of the general cleanliness of critical food-contact surfaces, walls, ceilings, and the identification of active contamination sources.
4. **Equipment and Furniture:** Maintenance logs, cleanliness, and the integrity of food-processing hardware.
5. **Storage Protocols:** Validation of cross-contamination prevention, adherence to refrigeration standards, and protection against pest infestation.
6. **Water Supply Standards:** Verification of access to an ample, continuous supply of potable water, and the sanitary condition of storage facilities.
7. **Waste Control Mechanisms:** Efficacy of solid waste collection, presence of closed receptacles, and the sanitary condition of adequate toilet facilities.
8. **Food Item Assessment:** Visual and documented inspection of packaging integrity, expiration date compliance, and exposure.
9. **Health Status of Handlers:** Verification of the mandated Schedule VI medical examinations, alongside visual confirmation of personal hygiene practices and the utilization of protective clothing.
10. **Training and Certification:** Documentation proving that food handlers have received formal instruction in Good Hygiene Practices (GHP) and Good Manufacturing Practices (GMP).

The Laravel backend will execute a mathematical aggregation of the scores inputted across these ten cages, calculating a final percentage. This percentage algorithmically dictates the grade assigned to the premises: Grade A (70–100%), Grade B (40–69%), or Grade C (0–39%).

The assigned grade is not merely cosmetic; it acts as an operational trigger within the life cycle management module. Grade A establishments are automatically scheduled for a follow-up inspection in six months, Grade B in three months, and Grade C mandates a highly scrutinized monthly inspection rhythm, which the system will automatically place into the PHI's future task queue.

3.4 Comprehensive Use Case Narratives

The following use case models explicitly detail the functional requirements, pre-conditions, and alternative flows required to develop the Laravel application. These narratives translate the flow diagram into precise software engineering directives.

Use Case 01: FBO Registration and Temporary Profile Creation

Description: The process by which an FBO establishes digital identity and creates a draft registration application.

Primary Actor: Food Business Operator (FBO).

Pre-conditions: The applicant possesses a valid National Identity Card (NIC) or Business Registration Number.

Post-conditions: A Temporary Profile is created in the database, and an application payload is securely stored in a DRAFT state.

Step	Action Flow
1.	The FBO navigates to the public FCAU IIMS portal and selects "Sign up to the system."
2.	The system authenticates the user via an email sent to a registered email address.
3.	The system generates a "Temporary Profile" database record.
4.	The FBO selects "Login to the Profile" and initiates the "Fill the Registration Form" interface.
5.	The FBO inputs biographic data, trade category, structural dimensions, and selects the geographical GN division.
6.	The system provides an encrypted upload portal for statutory documents (e.g., Trade License, Medical Certificates).
7.	The system auto-saves the payload incrementally, maintaining a DRAFT status to prevent data loss.

Alternative Flows:

- If the FBO attempts to upload a prohibited file type (e.g., .exe, compressed archives), the system rejects the file and throws an input validation error.

Use Case 02: Form Submission and Background Engine Validation

Description: The transition of the application from a draft state to a submitted state, triggering spatial and regulatory mapping.

Primary Actor: FBO, Automated Background Engines.

Pre-conditions: All mandatory fields in the Registration form are populated, and statutory documents are uploaded.

Post-conditions: The application transitions to SUBMITTED, mapped to a GN division and categorized.

Step	Action Flow
1.	The FBO clicks the "Submit the Form" trigger.
2.	The system executes the "Duplicate validation engine," querying the database for exact matches on NIC, Business Registration, and Address.
3.	The system executes the "GN Mapping" engine, utilizing the provided address to assign a specific GN spatial identifier.
4.	The system executes the "Premises categorization" engines, assigning the application to one of the 10 registered categories and calculating the baseline risk profile.
5.	The system transitions the entity state to SUBMITTED.

Alternative Flows:

- If the Duplicate validation engine detects a conflict, the submission is halted, and the FBO is prompted to contact the FCAU administrator or claim an existing business profile.

Use Case 03: MOH Application Routing and PHI Inspection (1st and 2nd Visits)

Description: The routing of the validated application to the appropriate health officials and the execution of physical field inspections.

Primary Actor: Medical Officer of Health (MOH), Public Health Inspector (PHI).

Pre-conditions: The application is SUBMITTED and linked to a GN division.

Post-conditions: The application is evaluated using the H-800 framework and either sent for final MOH approval or pushed to the appellate workflow.

Step	Action Flow
1.	The MOH logs into the dashboard and accesses the queue via the "Get the Application based on GN division assigned" function.
2.	The MOH reviews the application and selects "Send to PHI."
3.	The PHI accesses the application queue specific to their GN jurisdiction, transitioning the state to REVIEW_STATUS.
4.	The PHI conducts the "1st Visit" and inputs the 10 cages of the H-800 grading checklist into the mobile-responsive portal.
5.	If the aggregate score warrants an "Approve" decision, the system routes the application back to the MOH.
6.	If the score warrants a "Reject" decision, the system places the application in a PENDING_2ND_VISIT state and initiates a remediation timer.
7.	The PHI conducts the "2nd Visit" post-remediation. If approved, it routes to the MOH. If rejected, the system routes to a terminal state, triggering immediate dual notifications to the MOH and the Applicant.

Use Case 04: The Appellate Workflow and Profile Finalization

Description: The complex routing of a rejected application through the appellate structure, culminating in the creation of a permanent premises profile.

Primary Actor: FBO, Appeal Board, Regional Director (RD), MOH.

Pre-conditions: The application has suffered a final rejection post-2nd Visit.

Post-conditions: The application is conclusively rejected or transformed into an approved premises profile.

Step	Action Flow
1.	The FBO initiates an "Appeal" through the applicant portal.
2.	The system triggers the "Assign Appeal Board" subroutine, notifying designated administrative personnel.
3.	The Appeal Board records their "Approve / Reject" recommendation, and the system executes the "Send to RD" action.
4.	The RD accesses the "RD receives Appeal" interface, evaluates the board's findings, and executes a "Review and Approve" function.
5.	At the "Final Approval", if the RD approves, the state is forced to an override approval. If rejected, the rejection is sustained.
6.	The system executes the "MOH receives RD's Decision" logic. If approved, the MOH executes "Approve the application."
7.	The system finalizes the workflow by triggering the "Create the premises profile" module, deleting the temporary flag, and generating the record for the premises.

3.5 Data Dictionary and Entity-Relationship Architecture

The Laravel backend requires a highly normalized, relational database architecture (MySQL) to support the complex routing, GN mapping, and exhaustive audit trails required as per software standards. The following tables delineate the core schema required to support the data flow.

Entity: users (Authentication and Role Management)

This table manages the identity and Role-Based Access Control (RBAC) parameters for all actors as above.

Field Name	Data Type	Constraints	Description
id	BIGINT (Unsigned)	PRIMARY KEY	Unique system identifier.
nic_number	VARCHAR(12)	UNIQUE, NOT NULL	National Identity Card number used for primary verification.
full_name	VARCHAR(255)	NOT NULL	Statutory legal name of the user.
email_address	VARCHAR(255)	UNIQUE	Contact email for system notifications.
password_hash	VARCHAR(255)	NOT NULL	SHA256 or bcrypt or similar hashed authentication string.
role_type	ENUM	NOT NULL	Defined roles: 'FBO', 'PHI', 'MOH', 'APPEAL_BOARD', 'RD', 'ADMIN'.
assigned_gn_id	INT (Unsigned)	FOREIGN KEY	Maps PHIs and MOHs to specific spatial jurisdictions.
status	BOOLEAN	DEFAULT TRUE	Account activation toggle.

Entity: premises_applications (The Core Registration Payload)

This table maintains the state machine of the application as it flows from Draft to Approval or Rejection.

Field Name	Data Type	Constraints	Description
id	BIGINT (Unsigned)	PRIMARY KEY	Unique application dossier identifier.
fbo_user_id	BIGINT (Unsigned)	FOREIGN KEY	Links the application to the specific FBO.
business_name	VARCHAR(255)	NOT NULL	The trading name of the food establishment.
br_number	VARCHAR(50)	UNIQUE	Business Registration Number utilized by the Duplicate Engine.
category_rule_id	INT (Unsigned)	FOREIGN KEY	Maps to the "Premises Categorization - Regulation" engine output (1 of 10 types).
gn_division_id	INT (Unsigned)	FOREIGN KEY	Output of the "GN Mapping" engine, dictating workflow routing.
risk_category	ENUM	NOT NULL	Output of the "Premises Categorization - Risk-based" engine (High/Med/Low).
current_state	ENUM	NOT NULL	E.g., 'DRAFT', 'SUBMITTED', 'REVIEW_STATUS', '1ST_VISIT', 'APPEAL'.
is_temporary	BOOLEAN	DEFAULT TRUE	Remains TRUE until the "Create the premises profile" node is executed.
created_at	TIMESTAMP	AUTO	Timestamp of initial Temporary Profile creation.

Entity: h800_inspections (Field Grading Data)

This table captures the granular evaluation data generated by the PHI during the 1st and 2nd visits.

Field Name	Data Type	Constraints	Description
id	BIGINT (Unsigned)	PRIMARY KEY	Unique inspection record identifier.
application_id	BIGINT (Unsigned)	FOREIGN KEY	Links to the specific premises_applications record.
phi_user_id	BIGINT (Unsigned)	FOREIGN KEY	The Authorized Officer conducting the physical inspection.
visit_type	ENUM	NOT NULL	Differentiates between '1ST_VISIT' and '2ND_VISIT'.
cage_1_location	DECIMAL(5,2)	CHECK (0-10)	Rating for Location and Environment.
cage_2_building	DECIMAL(5,2)	CHECK (0-10)	Rating for Building construction and ventilation.
cage_3_process	DECIMAL(5,2)	CHECK (0-10)	Rating for Processing room cleanliness.
cage_4_equip	DECIMAL(5,2)	CHECK (0-10)	Rating for Equipment and furniture integrity.
cage_5_storage	DECIMAL(5,2)	CHECK (0-10)	Rating for Storage of foods protocols.
cage_6_water	DECIMAL(5,2)	CHECK (0-10)	Rating for Water supply.
cage_7_waste	DECIMAL(5,2)	CHECK (0-10)	Rating for Waste control mechanisms.

cage_8_food	DECIMAL(5,2)	CHECK (0-10)	Rating for Food item exposure and labeling.
cage_9_health	DECIMAL(5,2)	CHECK (0-10)	Rating for Health status of handlers (Schedule VI).
cage_10_training	DECIMAL(5,2)	CHECK (0-10)	Rating for Training received by food handlers.
total_percentage	DECIMAL(5,2)	NOT NULL	Calculated aggregate score dictating the Grade.
assigned_grade	ENUM('A','B','C')	NOT NULL	Algorithmic outcome (A: 70-100%, B: 40-69%, C: 0-39%). ⁷
decision	ENUM('APPROVE','REJECT')	NOT NULL	Boolean trigger advancing the workflow state machine.

Entity: appeal_proceedings (Grievance Tracking)

This table manages the complex multi-actor workflow triggered upon a terminal inspection rejection.

Field Name	Data Type	Constraints	Description
id	BIGINT (Unsigned)	PRIMARY KEY	Unique identifier for the appeal dossier.
application_id	BIGINT (Unsigned)	FOREIGN KEY	Links to the rejected application.
appeal_board_id	BIGINT (Unsigned)	FOREIGN KEY	Links to the specific assigned committee identifier.
board_decision	ENUM('APPROVE','REJECT')	NULLABLE	The preliminary decision formulated by the Appeal Board.
rd_user_id	BIGINT (Unsigned)	FOREIGN KEY	Links to the specific Regional Director handling the final tier.
rd_final_decision	ENUM('APPROVE','REJECT')	NULLABLE	The "Final Approval" diamond outcome dictating MOH notification.
appeal_filed_at	TIMESTAMP	NOT NULL	Date the PO initiated the grievance protocol.
resolved_at	TIMESTAMP	NULLABLE	Timestamp of the final RD decision execution.

Entity: compliance_documents (File Artifact Management)

This table maintains pointers to the securely stored, malware-scanned documents uploaded by the FBO.

Field Name	Data Type	Constraints	Description
id	BIGINT (Unsigned)	PRIMARY KEY	Unique document identifier.
application_id	BIGINT (Unsigned)	FOREIGN KEY	Links to the parent application payload.
document_type	VARCHAR(100)	NOT NULL	E.g., 'BR_CERTIFICATE', 'SCHEDULE_VI_MEDICAL', 'TRADE_LICENSE'.
storage_path	VARCHAR(512)	NOT NULL	Immutable path to the file in the Cloud object storage.
file_hash	VARCHAR(256)	NOT NULL	SHA-256 or similar hash to verify file integrity post-upload.
malware_status	ENUM	DEFAULT 'PENDING'	Tracks the queue state of the asynchronous malware scan.

4. Inspection and Grading Module

The primary objective of the Inspection and Grading Module is to digitize the end-to-end workflow of food premises registration, on-site hygiene inspections, supervisory reviews, and the subsequent grading and certification processes. Historically reliant on fragmented, paper-based instruments such as the Schedule I application form and the Health-800 (H800) inspection checklist, the transition to a centralized digital architecture will eliminate data silos, ensure real-time regulatory oversight.

The system orchestrates interactions between four primary actors: the Food Control Administration Unit (FCAU) acting as the central configurator; the Public Health Inspector (PHI) acting as the field executioner; the Medical Officer of Health (MOH) acting as the supervisory reviewer; and the FBO acting as the regulated entity. By standardizing the evaluation algorithms across these actors, tracking compliance branches, and automating the issuance of advisory reports and certificates, the module will introduce unprecedented transparency, traceability, and efficiency into the Sri Lankan food control supply chain.

4.1 The H800 Inspection and Grading Algorithm

The architectural logic and business rules of the Inspection and Grading Module are inextricably linked to the statutory instruments enforced by the FCAU.

Upon the successful routing of a scheduled inspection, the PHI is mandated to inspect the premises using the primary regulatory instrument: the Grading of Food Handling Establishments form, universally known as the H800 (revised 2022). The H800 logic is sophisticated and requires dynamic algorithmic adjustments within the software based on the establishment category and the specific operations conducted on-site.

The assessment is divided into key domains: General, Building, Food Handler, Processing and/or Serving Area, and Food Storage. The scoring engine must implement variable scoring matrices. Items are typically scored on a 0, 2, 4, or 5-point scale based on the degree of compliance. For instance, the provision of "Safe Water" is scored as 0 (no water), 2 (water available but not safe), or 5 (safe water available and utilized correctly). Similarly, "Waste Water Disposal" is scored as 0 (insanitary disposal directly to open ground) or 2 (sanitary disposal via septic tanks or treatment plants).

The software must algorithmically adjust the maximum possible score, referred to as the denominator, based on the specific context of the establishment. If a facility has no processing or serving area, such as a packaged grocery store, 20 marks are automatically deducted from

the total possible denominator. If there is no storage of cooked food, 16 marks are deducted. The final percentage is mathematically derived by dividing the total achieved score by the adjusted denominator.

The system must automatically assign a statutory grade based on the calculated percentage thresholds. Grade A, denoting a "Good" establishment, requires a score between 75% and 100%. Grade B, denoting "Satisfactory," requires a score between 50% and 74%. Grade C, denoting "Unsatisfactory," requires a score between 25% and 49%. Grade D, denoting "Very Unsatisfactory," is assigned for scores between 0% and 24%. To prevent an establishment from masking critical failures with high scores in trivial areas, the system must enforce mandatory minimums; to achieve an A or B rating, the establishment must obtain at least 75% or 50%, respectively, across all relevant individual sections.

4.2 Comprehensive Workflow Elaboration

This section elaborates on the precise sequence of operations required by the software architecture.

Phase 1: FCAU Configuration and Parameterization

The workflow commences at the apex of the regulatory hierarchy with the Food Control Administration Unit (FCAU). The system must provide a dedicated administrative interface allowing the FCAU actor to configure the foundational parameters of the grading module. The FCAU provides three primary data inputs into the system. First, "**Food Classification**," which dictates the categorization matrices for different types of edible goods and their inherent risk profiles. Second, the "**Grading criteria under each classification**," which involves mapping the specific H800 checklist items, their respective weights, and the dynamic denominator rules to the specific food classifications. Third, the "**Inspection rules and frequency**," which defines the temporal algorithms for how often an establishment must be visited based on its historical performance or base category. These three inputs converge into a central system process node designated as "Configure Grading Form."

Phase 2: PHI Calendar Generation and Trigger Mechanisms

The field execution phase, managed by the Public Health Inspector (PHI), relies on a sophisticated task routing engine. The PHI does not arbitrarily select premises for inspection; rather, the system populates their queue through three distinct event triggers. The first is a "Risk based trigger," an automated systemic event generated when an establishment's routine inspection interval has elapsed (e.g., three months for a Grade A facility, one month for a Grade

C facility). The second trigger is a "Complaint," originating from consumer portals or other health authorities indicating a suspected hygiene violation or foodborne illness link. The third trigger is a "Follow up," manually or automatically generated to verify that previously issued corrective actions have been implemented.

These three triggers feed into the PHI's primary interface process: "Get inspection Calendar view Based on assigned GN." The system must filter and aggregate these inspection tasks spatially, presenting the PHI with a structured calendar and geographical mapping organized by their assigned Grama Niladhari (GN) divisions. This allows for optimized route planning and efficient deployment of public health resources.

Phase 3: Field Inspection Execution

From the GN-based calendar view, the PHI initiates the "Schedule inspection" process, locking the task and notifying relevant stakeholders. Upon arrival at the physical premises, the PHI interacts with the module to execute the "Record inspection date/time" function, securely logging the temporal and geospatial metadata of the visit to prevent fraudulent reporting.

The core of the PHI's field activity is the "Complete grading form" process. Here, the application renders the configured H800 checklist. The PHI systematically evaluates the premises against the standardized criteria. Simultaneously, the system must support the "Attach support docs/photos" function. This allows the PHI to use the mobile device's camera to capture forensic evidence of both compliance and non-compliance; such as photographing a malfunctioning refrigeration unit or an unsanitary waste disposal area and immutably linking these digital artifacts directly to the specific grading criteria line item.

Phase 4: Automated Grading and Submission

Once the PHI completes the data entry for the checklist, the system transitions to the "Auto generated grading calculation" node. The software engine applies the FCAU configured algorithms, processing the dynamic denominators and aggregating the scores to determine the final percentage and corresponding statutory grade (A, B, C, or D).

Following the calculation, the system executes the "Record violations and non-compliance findings" process. This automated sub routine scans the completed checklist for any criteria that received sub optimal scores (e.g., scores of 0 or 2 on a 5-point scale). It aggregates these specific failures into a formal digital document, which details the exact regulatory breaches. The PHI then executes the "Submit for review" process, securely transmitting the entire encrypted

inspection payload including the raw scores, the calculated grade, the photographic evidence, and the recorded violations to the central server for supervisory oversight.

Phase 5: MOH Supervisory Review and Decision Branching

The workflow then transitions to the Medical Officer of Health (MOH), who serves as the supervisory authority. The MOH accesses a centralized dashboard to execute the "Review inspection" process. The MOH examines the PHI's submitted payload, verifying the photographic evidence against the recorded scores and ensuring that the automated grade aligns with the qualitative reality of the premises.

This review leads to a critical decision node: "Review." If the MOH identifies discrepancies, missing evidence, or logical errors in the PHI's report, they trigger the "Return for corrections" path. This action reroutes the workflow back to the PHI at the "Record violations and non-compliance findings" stage, requiring the PHI to amend the documentation, gather additional evidence, or clarify remarks before resubmitting.

Conversely, if the MOH validates the integrity and accuracy of the inspection, they proceed to the "Approve" process, cryptographically signing the inspection record and locking it into the immutable database.

Phase 6: Compliance Routing and Enforcement

The MOH's approval immediately feeds into a secondary, automated decision node labeled "Compliance." The system evaluates the finalized grade and routes the workflow down one of three distinct paths based on the severity of the findings.

If the establishment achieves "High compliance" (typically Grade A or B), the system loops the workflow directly back to the PHI's "Get inspection Calendar view" node, effectively placing the establishment back into the routine, risk-based surveillance queue for a future date (e.g., two to three months later).

If the establishment demonstrates "Moderate" compliance (typically Grade C), the system routes the workflow to the "Corrective Actions" process. This generates formal, time bound mandates requiring the Premises Owner to rectify specific structural or operational deficiencies.

If the establishment exhibits "Low" compliance (typically Grade D or catastrophic critical failures), the system routes the workflow to the critical "Enforcement/Follow-up" node. This triggers severe administrative or legal actions, which may include the immediate suspension of

the trade license, the initiation of food seizure protocols, or the preparation of documents for judicial prosecution under the Food Act No. 26 of 1980.

Parallel to these internal regulatory routings, the system executes the final process for the FBO actor: "Receive results, get the grading." The system automatically transmits the finalized grade, the compliance certificate, and any associated corrective action mandates or enforcement notices to the FBO via a secure system and email notifications.

4.3 Use Case Modeling and Narratives

To translate the architectural flow diagram into actionable software development parameters, the system's functionalities are modeled through detailed use case narratives. These narratives define the exact system behaviors, preconditions, and alternate flows required to support the regulatory mandate.

Use Case 01: Configure Grading Form

Description: The FCAU administrator configures the foundational rules, risk categorizations, grading criteria matrices (H800 checklist items and weights), and inspection frequency algorithms.

Primary Actor: FCAU

Pre Conditions: The FCAU administrator is authenticated with paramount system privileges and is accessing the configuration dashboard.

Post Conditions: Grading configurations are securely saved in the database, versioned, and made available for synchronization with the PHI functions.

Step	Action Flow
1	FCAU defines and inputs "Food Classification" parameters into the system.
2	FCAU maps the "Grading criteria under each classification", adjusting dynamic denominators and point scales.
3	FCAU configures the "Inspection rules and frequency" based on establishment types and risk profiles.
4	System executes the "Configure Grading Form" node, validating the business logic of the combined inputs.
5	System commits the configuration to the central database and updates the active version.

Use Case 02: Generate Inspection Calendar

Description: The system aggregates automated and manual triggers to generate a spatially organized inspection calendar for the Public Health Inspector based on their assigned Grama Niladhari (GN) division.

Primary Actor: PHI (assisted by System)

Pre Conditions: The PHI is authenticated and assigned to specific GN divisions. FCAU configuration parameters are active.

Post Conditions: The PHI views a populated, actionable calendar and successfully schedules targeted premises for inspection.

Step	Action Flow
1	System continuously monitors and aggregates "Risk based trigger" events (expired inspection intervals).
2	System aggregates logged "Complaint" events tied to registered premises.
3	System aggregates "Follow up" events triggered by previous moderate compliance results.
4	PHI accesses the system and requests to "Get inspection Calendar view Based on assigned GN".
5	System filters the aggregated triggers against the PHI's GN and renders the calendar/map interface.
6	PHI selects an establishment from the calendar view and executes the "Schedule inspection" function.

Use Case 03: Execute Field Inspection

Description: The PHI conducts an on-site evaluation of the scheduled premises, records temporal data, inputs scores using the H800 checklist configurable by the FACU admin, and attaches digital forensic.

Primary Actor: PHI

Pre Conditions: An inspection task is scheduled. The PHI has arrived at the physical premises and possesses a synchronized mobile device to access the system.

Post Conditions: Raw inspection scores and encrypted digital evidence are securely saved in the cloud database.

Step	Action Flow
1	PHI accesses the scheduled task and executes "Record inspection date/time".
2	System automatically logs the current timestamp and PHI to select GPS coordinates.
3	System renders the dynamic H800 checklist based on the establishment's classification.
4	PHI navigates the checklist domains and executes "Complete grading form" by inputting observed scores (e.g., 0, 2, 4, 5).
5	For specific observations (compliance or non-compliance), PHI executes "Attach support docs/photos".
6	System captures the image, and links the artifact to the specific checklist item.

Use Case 04: Process Automated Grading

Description: The system's rules engine processes the raw checklist data, applies dynamic denominator adjustments, calculates the final percentage, and determines the statutory grade.

Primary Actor: System

Pre Conditions: The PHI has completed the data entry for the grading form.

Post Conditions: A final percentage score and corresponding grade (A, B, C, or D) are calculated and appended to the inspection record.¹

Step	Action Flow
1	System initiates the "Auto generated grading calculation" process upon form completion.
2	System evaluates the establishment type and deducts non-applicable marks from the base denominator.
3	System aggregates all marks awarded by the PHI across the checklist.
4	System calculates the final percentage (Achieved Score / Adjusted Denominator * 100).
5	System evaluates the percentage against statutory thresholds and mandatory minimum rules.
6	System assigns the finalized statutory grade (A, B, C, or D) to the record.

Use Case 05: Record Violations and Submit

Description: The system isolates non-compliant checklist items, allows the PHI to generate an advisory report, and securely transmits the entire inspection payload to the MOH for review.

Primary Actor: PHI (assisted by System)

Pre Conditions: Automated grading is complete, or the task has been returned for corrections by the MOH.

Post Conditions: The finalized inspection payload is encrypted and placed in the MOH's pending review queue.

Step	Action Flow
1	System executes "Record violations and non-compliance findings" by scanning the checklist for sub-optimal scores.
2	System compiles identified failures into a structured digital advisory report format.
3	PHI reviews the compiled violations and appends necessary technical remarks or citations.
4	PHI verifies the complete inspection dossier (scores, photos, violations).
5	PHI executes the "Submit for review" function.
6	System encrypts the payload and routes it to the MOH interface.

Use Case 06: Supervisory Review

Description: The Medical Officer of Health evaluates the PHI's submitted inspection dossier to ensure accuracy and procedural compliance before making a final approval or rejection decision.

Primary Actor: MOH

Pre Conditions: A completed inspection payload is available in the MOH's review queue.

Post Conditions: The inspection is either permanently approved and locked, or returned to the PHI's queue for required amendments.

Step	Action Flow
1	MOH accesses the dashboard and initiates the "Review inspection" process.
2	MOH evaluates the scores, attached photographic evidence, and recorded violations.
3	MOH reaches the "Review" decision node.
4	If discrepancies exist, MOH selects the "Return for corrections" path.
5	System routes the workflow back to the PHI at the "Record violations and non-compliance findings" node.
6	If the dossier is accurate, MOH selects the "Approve" path.
7	System applies an e-signature and locks the inspection record as an immutable artifact.

Use Case 07: Manage Compliance Routing

Description: Following MOH approval, the system evaluates the final grade and automatically routes the workflow to the appropriate subsequent regulatory action path.

Primary Actor: System

Pre Conditions: The inspection has been approved by the MOH.

Post Conditions: The establishment is queued for future routine inspection, issued corrective action mandates, or flagged for legal enforcement.

Step	Action Flow
1	System evaluates the approved grade at the "Compliance" decision node.
2	If grade indicates "High compliance", system routes to the "High compliance" loop back.
3	System injects a new "Risk based trigger" into the future calendar queue based on frequency rules.
4	If grade indicates "Moderate" compliance, system routes to the "Corrective Actions" node.
5	System generates time-bound remediation mandates and schedules a "Follow up" trigger.

6	If grade indicates "Low" compliance, system routes to the "Enforcement/Follow-up" node.
7	System generates alerts for potential license suspension or legal prosecution procedures.

Use Case 08: Issue Grading Results

Description: The system generates the final compliance documentation and securely transmits the results, grades, and associated mandates to the regulated Premises Owner.

Primary Actor: System (interacting with FBO)

Pre Conditions: The compliance routing decision has been finalized by the system.

Post Conditions: The FBO possesses a digital Certificate of Compliance and any relevant advisory or enforcement notices.

Step	Action Flow
1	System generates a digital Certificate of Compliance containing the finalized grade and a verification QR code.
2	System compiles the violation report and any specific "Corrective Actions" or "Enforcement/Follow-up" documentation.
3	System executes the "Receive results, get the grading" process for the Premises Owner actor.
4	System transmits the compiled documentation package via email and/or the secure web portal.
5	FBO downloads and reviews the grading and required actions.

4.4 Data Dictionary and Entity-Relationship Architecture

To support the complex legal, procedural, and algorithmic nuances of the FCAU grading mechanisms, the relational database must be rigorously structured. The architecture must accommodate dynamic configurations, massive geospatial datasets, and immutable forensic auditing. The following tables outline the core entity-relationship architecture required to actualize the flow diagram.

Entity: master_configuration (Configuration)

This table stores the foundational rules dictated by the FCAU actor during the "Configure Grading Form" process.

Field Name	Data Type	Constraint	Description
config_id	UUID	Primary Key	Unique identifier for the configuration rule.
classification_code	VARCHAR(10)	Not Null	Code representing the food risk classification.
h800_criteria_id	VARCHAR(20)	Foreign Key	Links to specific line items in the grading checklist.
max_score_weight	INTEGER	Not Null	The maximum allowable score for this criteria (e.g., 2, 4, 5).
denominator_deduction	INTEGER	Default 0	Points to deduct from the base denominator if N/A.
inspection_frequency	INTEGER	Not Null	Days until the next risk-based trigger is generated.
active_status	BOOLEAN	Default TRUE	Indicates if the rule is currently enforced.

Entity: inspection_trigger

This table manages the task queue populated during the "Get inspection Calendar view" process.

Field Name	Data Type	Constraint	Description
trigger_id	UUID	Primary Key	Unique identifier for the scheduled event.
establishment_id	UUID	Foreign Key	Links to the physical premises in the registry.
trigger_type	ENUM	Not Null	Values: RISK_BASED, COMPLAINT, FOLLOW_UP.
gn_division_code	VARCHAR(50)	Not Null	Geographical boundary used for PHI routing.
target_date	DATE	Not Null	The deadline by which the inspection must occur.
phi_assigned_id	UUID	Foreign Key	Links to the specific PHI tasked with the inspection.
status	ENUM	Not Null	Values: PENDING, SCHEDULED, IN_PROGRESS, COMPLETED.

Entity: h800_inspection_master

This table stores the metadata and finalized calculations for a specific field inspection event.

Field Name	Data Type	Constraint	Description
inspection_id	UUID	Primary Key	Unique identifier for the inspection execution.
trigger_id	UUID	Foreign Key	Links back to the event that initiated the inspection.
execution_timestamp	TIMESTAMP	Not Null	Exact timestamp recorded during the field visit.
gps_latitude	DECIMAL(10,8)	Not Null	Geospatial evidence of PHI location.
gps_longitude	DECIMAL(11,8)	Not Null	Geospatial evidence of PHI location.
base_denominator	INTEGER	Default 100	The maximum possible score before dynamic adjustments.
adjusted_denominator	INTEGER	Not Null	Denominator post FCAU configuration adjustments.
total_score_achieved	INTEGER	Not Null	Sum of all marks awarded during the checklist process.
final_percentage	DECIMAL(5,2)	Not Null	Computed value determining the grade.
calculated_grade	ENUM	Not Null	Values: A, B, C, D based on threshold mapping.
moh_approval_status	ENUM	Not Null	Values: PENDING, RETURNED, APPROVED.

Entity: h800_line_items

This table captures the granular data entry performed during the "Complete grading form" process, tracking specific hygienic evaluations.

Field Name	Data Type	Constraint	Description
item_record_id	UUID	Primary Key	Unique identifier for the specific observation.
inspection_id	UUID	Foreign Key	Links to the master inspection record.
criteria_id	VARCHAR(20)	Foreign Key	E.g., 'BLD-01' (Safe Water), 'STG-04' (Waste Bins).
awarded_score	INTEGER	Not Null	The score input by the PHI (e.g., 0, 2, 4, 5).
is_not_applicable	BOOLEAN	Default FALSE	True if the item was bypassed based on establishment type.
violation_flag	BOOLEAN	Not Null	Auto-set to TRUE if the score is sub-optimal.

Entity: inspection_evidence

This table securely links the digital artifacts captured during the "Attach support docs/photos" process.

Field Name	Data Type	Constraint	Description
evidence_id	UUID	Primary Key	Unique identifier for the digital artifact.
item_record_id	UUID	Foreign Key	Links the photo directly to a specific checklist score.
media_type	ENUM	Not Null	Values: PHOTO, DOCUMENT, SIGNATURE_BLOB.
storage_uri	VARCHAR(512)	Not Null	Secure path to encrypted cloud blob storage.

capture_timestamp	TIMESTAMP	Not Null	Immutable timestamp of when the evidence was acquired.
cryptographic_hash	VARCHAR(256)	Not Null	SHA-256 or similar hash ensuring the photo has not been tampered with.

Entity: compliance_routing

This table tracks the outcomes of the MOH review and the subsequent compliance branching logic.

Field Name	Data Type	Constraint	Description
routing_id	UUID	Primary Key	Unique identifier for the compliance action.
inspection_id	UUID	Foreign Key	Links to the approved master inspection.
compliance_level	ENUM	Not Null	Values: HIGH, MODERATE, LOW.
action_taken	ENUM	Not Null	Values: ROUTINE_LOOP, CORRECTIVE_MANDATE, ENFORCEMENT.
advisory_report_uri	VARCHAR(512)	Nullable	Link to the generated report.
certificate_qr_hash	VARCHAR(256)	Nullable	The cryptographic string embedded in the issued certificate.

5. Laboratory Coordination Module

The Lab Coordination Module constitutes the third core component of the food safety inspection management system. Under the Food Act No. 26 of 1980, the analytical verification of food samples is a critical legal procedure used to prevent fraud, ensure standard compliance, and protect public health.

This module bridges the gap between field enforcement and scientific verification. It digitizes the chain of custody, starting from the moment a Public Health Inspector (PHI) or Food and Drug Inspector (FDI) dispatches a physical sample, through the laboratory triage and testing phases, and concluding with the ingestion of analytical reports back into the central regulatory pdfsystem. The module accommodates the transitional state of Sri Lankan laboratories by supporting both automated API integrations for modern Laboratory Information Management Systems (LIMS) and intelligent PDF parsing for legacy manual reporting workflows.

5.1 Comprehensive Workflow Elaboration

The system orchestrates a multi-actor workflow involving the PHI/FDI, the Laboratory Reception Counter, the Approved Analyst, and the core System engine.

Phase 1: Lab Classification & Sample Dispatch

The workflow begins with "Lab Classification," an administrative configuration defining the testing capabilities and authorization status of various labs. In the field, the PHI or FDI executes the "Send samples to Lab" process. They log the sample metadata (source, timestamp, required parameters) into the system and physically transport the sealed sample to the designated facility.

Phase 2: Reception Triage

Upon arrival, the sample reaches the "Reception Counter" actor. The receptionist evaluates the physical package against the digital metadata to ensure integrity (e.g., correct temperature, untampered wax seals, sufficient quantity). This is a decision node:

- Reject: If the sample is compromised, the system routes the workflow back to the PHI/FDI via the "Sends correct sample" path, mandating a recollection.

- Accept: If valid, the receptionist executes "Accept and send the sample to Approved Analyst," transferring custody internally.

Phase 3: Testing and Manual Reporting

The sample reaches the "Approved Analyst," a legally designated role. The analyst executes "Receive the samples" and proceeds with the "Lab testing process - Manual." Due to the physical nature of chemical and microbiological analysis, this step happens outside the software. Following testing (which takes 5 to 10 days depending on the parameter), the analyst executes "Generate Lab Report - Manual".

Phase 4: Data Ingestion (The Integration Node)

The system must capture the results of the manual report. The architecture provides two paths based on the "System to access the lab report" decision:

- API Path: "if APIs are available," the system runs a scheduled cron job to "Fetch data through APIs to the system" directly from the laboratory's internal LIMS.
- PDF Parsing Path: If no API exists, the Approved Analyst must "Manually upload the report to the system in PDF format." Once uploaded, the system utilizes OCR and data-mining algorithms to "read the data from PDF and fetch it" into structured database fields.

Phase 5: Registry and Dispatch

Both ingestion paths converge at the "Maintains a register" node. The system links the extracted results to the system-generated sample number. Finally, the system executes "Send reports to MOH," instantly notifying the field officer of the results so they can initiate enforcement or clearance actions.

5.2 Use Case Modeling and Narratives

Use Case 1: Send Samples to Lab

Description: The field officer logs the collection of a physical sample and digitally dispatches the requisition to the appropriate laboratory.

Primary Actor: PHI / FDI

Pre Conditions: A physical sample has been legally collected and divided into three parts.

Post Conditions: A digital requisition is created, and the sample is marked as 'In Transit'.

Step	Action Flow
1	PHI/FDI selects the target "Lab Classification" from the system registry.
2	PHI/FDI inputs sample metadata (quantity, container type, required chemical/microbiological tests).
3	System generates a unique tracking QR/Sample ID.
4	PHI/FDI physically dispatches the sample and executes the "Send samples to Lab" function in the system.

Use Case 2: Triage and Accept/Reject Sample

Description: The lab reception verifies the integrity of the incoming physical sample against the digital requisition.

Primary Actor: Reception Counter

Pre Conditions: The physical sample arrives at the lab.

Post Conditions: Custody is transferred to the Approved Analyst, or the sample is rejected back to the PHI.

Step	Action Flow
1	Reception Counter scans the sample QR/ID to pull up the digital requisition.
2	Receptionist evaluates the physical integrity (temperature, seals, minimum weight).

3	If compromised, Receptionist triggers the "Reject" path. System notifies PHI to "Sends correct sample".
4	If valid, Receptionist triggers the "Accept" path.
5	System logs the custody transfer and executes "Accept and send the sample to Approved Analyst".

Use Case 3: Ingest Report via API

Description: The system automatically pulls structured analytical results from an interconnected laboratory system.

Primary Actor: System

Pre Conditions: The Approved Analyst has completed the "Generate Lab Report - Manual" step in an API-enabled lab.

Post Conditions: Analytical data is structured and saved within the central registry.

Step	Action Flow
1	System evaluates the integration capabilities of the assigned lab.
2	System determines "if APIs are available".
3	System initiates a secure REST request to the laboratory's endpoint using the Sample ID.
4	System executes "Fetch data through APIs to the system", pulling JSON/XML payloads.
5	System maps the payload to the internal database structure.

Use Case 4: Upload and Parse PDF Report

Description: For labs without APIs, the analyst uploads a document, and the system extracts the necessary data using parsing logic.

Primary Actor: Approved Analyst / System

Pre Conditions: The Approved Analyst has completed the "Generate Lab Report - Manual" step.

Post Conditions: The PDF is securely archived, and its contents are extracted into the central registry.

Step	Action Flow
1	Approved Analyst accesses the portal and executes "Manually upload the report to the system in PDF format".
2	System stores the encrypted PDF blob in the secure file repository.
3	System executes background workers to "read the data from PDF and fetch it".
4	System uses OCR and positional mapping to extract parameters (e.g., E. coli counts, Heavy metal presence).
5	System flags any parsing anomalies for manual review.

Use Case 5: Maintain Registry and Dispatch Results

Description: The system updates the master ledger and alerts the originating field officer of the outcome.

Primary Actor: System

Pre Conditions: Data ingestion (via API or PDF) is successfully completed.

Post Conditions: The MOH possesses the final report for regulatory action.

Step	Action Flow
1	System executes "Maintains a register", merging the ingested analytical data with the original "Sample number generated by system".
2	System checks the results against statutory thresholds.
3	System packages the final dossier, including the original requisition and the lab's findings.
4	System executes "Send reports to MOH" via push notification and dashboard alert.

5.3 Data Dictionary and Entity-Relationship Architecture

To support the chain of custody and data parsing requirements, the database requires the following core entities:

Entity: lab_classification

Field Name	Data Type	Constraint	Description
lab_id	UUID	Primary Key	Unique identifier for the laboratory.
lab_name	VARCHAR	Not Null	Name of the facility (e.g., Govt Analyst Dept).
is_authorized	BOOLEAN	Default FALSE	True if authorized under the Food Act for formal samples.
api_enabled	BOOLEAN	Default FALSE	Determines the reporting flow (API vs PDF upload).
api_endpoint	VARCHAR	Nullable	Base URL for RESTful data fetching if applicable.

Entity: sample_registery

Field Name	Data Type	Constraint	Description
sample_id	UUID	Primary Key	System-generated unique tracking number.
officer_id	UUID	Foreign Key	Links to the PHI/FDI who collected the sample.
food_category	VARCHAR	Not Null	Type of food (e.g., Spices, Water, Milk Powder).
quantity_collected	DECIMAL	Not Null	The weight/volume of the sample.
status	ENUM	Not Null	IN_TRANSIT, REJECTED, IN_LAB, COMPLETED.
rejection_reason	TEXT	Nullable	Populated if the Reception Counter rejects the sample.

Entity: analytical_reports

Field Name	Data Type	Constraint	Description
report_id	UUID	Primary Key	Unique identifier for the lab result.
sample_id	UUID	Foreign Key	Links back to the physical sample.
analyst_id	UUID	Foreign Key	Links to the Approved Analyst handling the test.
ingestion_method	ENUM	Not Null	Values: API_FETCH, PDF_PARSED.
raw_pdf_uri	VARCHAR	Nullable	Secure storage link to the manually uploaded PDF.
is_compliant	BOOLEAN	Not Null	True if parameters meet national standards.
dispatch_timestamp	TIMESTAMP	Nullable	Time the report was sent back to the PHI/FDI.

6. Legal, Prosecution and Enforcement Management Module

The primary mandate of the FCAU is to ensure the safety, hygiene, and authentic representation of the domestic and imported food supply chain. Historically, regulatory enforcement in this domain has relied heavily on localized, end product testing and manual record keeping by field officers, which often limits the capacity for systemic risk based surveillance. By digitizing the legal and enforcement workflows, this module seeks to eradicate the inefficiencies inherent in the manual Public Health Inspector (PHI) Monthly Returns, strictly enforce statutory timelines mandated, and provide a unified, highly auditable chain of custody for all regulatory actions.

6.1 Comprehensive Workflow Elaboration

The core of the Legal, Prosecution, and Enforcement Management Module is a master state-machine workflow that synthesizes multiple regulatory pathways into a single, cohesive digital journey.

Phase 01: Initiation and Input Parameters

The workflow is triggered by an Authorized Officer (PHI) operating within the field or at a designated port of entry. The system requires the ingestion of critical contextual metadata before a formal action can be logged. As defined by the input architectures, the officer must select parameters from four primary data arrays:

1. **Food Classification:** Identifying the specific commodity (e.g., dairy, spices, cereals, bottled water) to invoke the correct regulatory thresholds.
2. **Process Types:** Defining the context of the interaction. The system presents an enumerated list including: *Seized, Prosecuted, Arrested, H800 (Premises Inspection), Import, Food Sampled, Water Sampled and Salt Manufacturing*.
3. **Regulations Main and Sub Categories:** Mapping the observed condition to the specific statutory framework (e.g., Food Hygiene Regulations 2011, Preservatives Regulation 2019).
4. **Action Taken:** A preliminary declaration of the intended enforcement posture, primarily categorized as *Detained* or *Destroyed*.

The primary node of the workflow commences with the event source: "**From Inspection, Grading, Imports, Sampling, Enforcement / Follow-up**". This unified entry point ensures that regardless of whether a violation is detected during a routine H-800 restaurant grading, a border import sampling procedure, or a targeted surveillance follow-up, the subsequent legal tracking utilizes the same immutable chain of custody.

Phase 02: The Main Action Matrix

Once the event source and input parameters are logged, the workflow advances to the "**Main Action**" decision gate. The Authorized Officer must determine the immediate physical status of the food article or premises in question. The system forces a directional choice:

- **Released:** If the officer determines that the provisions of the Food Act and its regulations have not been contravened, or if an initial suspicion is cleared upon further inspection, the goods are released. The system logs the interaction for audit purposes, and the workflow terminates successfully.
- **Destroyed:** If the food is visibly putrid, decayed, or poses an imminent lethal threat, and the owner provides legally binding written consent for its destruction, the officer executes the destruction. The system generates the necessary destruction documentation, the inventory is neutralized, and the workflow terminates.
- **Detained:** If the food is suspected of adulteration requiring laboratory confirmation, or if it is held pending a court order due to an owner's refusal to consent to destruction, the goods are seized and detained under the Food Act.

Phase 03: The Detained Action Escalation

For goods routed through the "Detained" path, the system transitions to a secondary holding state pending external variables (e.g., Approved Analyst test results or judicial decrees). The workflow arrives at the "**Detained Action**" diamond, which periodically evaluates the status of the hold.

- If laboratory analysis exonerates the product, the status shifts to **Released**, and the workflow terminates.
- If a Magistrate issues an order for condemnation, or if the owner subsequently provides written consent, the status shifts to **Destroyed**, terminating that branch of the workflow.

- If the laboratory results confirm severe adulteration, or if the sanitary violations warrant punitive action beyond mere confiscation, the workflow escalates to the **Prosecuted** state.

Phase 04: The Prosecution Action Routing

When the workflow breaches the threshold for formal legal action; either directly from a severe initial inspection or escalated from a detained state, it enters the "**Prosecution Action**" decision matrix. This is the most legally sensitive node in the system. The software requires the Authorized Officer to explicitly define the legal foundation for the plaint. The system branches into three distinct pathways:

1. **Food Act:** The prosecution relies solely on the primary provisions and prohibitions of the Food Act
2. **Regulations:** The prosecution is based upon a violation of a specific subsidiary gazette, such as the failure to declare a batch number under the Labelling and Advertising Regulations, or operating without a permit under the Registration of Premises Regulations.
3. **Food Act & Regulations Both:** The prosecution is a composite charge, alleging violations of the principal act compounded by specific regulatory failures (e.g., manufacturing deleterious food under insanitary conditions *and* failing to adhere to mandatory color-coding standards).

Phase 05: Terminal State: The Prosecution Register

Regardless of which prosecution pathway is selected (Act, Regulations, or Both), all legal enforcement branches converge on the final mandatory node: "**Fill the Prosecution Register**".

This node represents the digitization of the traditional manual logbook. The system will not allow the workflow to be marked as 'Complete' until the officer inputs the generated case number, the specific court of jurisdiction, the calling dates, the details of the accused, and ultimately, the final judicial verdict and penalty details. Once the register is fully populated and signed off by a supervising Medical Officer of Health (MOH), the enforcement loop is officially closed, and the data is archived for predictive analytics and future risk profiling.

6.2 Use Case Modeling and Narratives

Use Case 1: Execute Seizure and Formulate Destruction Orders

Description: To provide a digital framework for the lawful seizure of hazardous goods under the Food Act and to manage the strict legal protocols distinguishing between consensual destruction and judicially ordered destruction

Primary Actor/s: Authorized Officer, FBO, System

Preconditions: The officer physically identifies food that is profoundly decayed, insect-infested, or unfit for consumption, requiring immediate removal from the supply chain.

Step	Action Flow
1	The officer selects the "Seizure" pathway, corresponding to the "Main Action" decision gate in the core workflow.
2	The system generates a digital Notice of Seizure. The officer inputs exact inventory counts, location specifics, and captures comprehensive photographic evidence of the putrefaction or infestation
3	The system presents a "Consent for Destruction" digital form
4	The officer logs the physical disposal method, uploads corroborating images of the destroyed goods, and the system terminates the workflow

6.3 Data Dictionary and Entity-Relationship Architecture

Entity: Prosecution_Registry

This entity replaces the physical "Notices and Prosecutions Register," serving as the definitive record of all legal actions initiated by the FCAU.

Field Name	Data Type	Constraint / Validation	Description
Case_Registry_No	String	Primary Key, Format: YYYY/Seq	Sequential identifier representing the legal case.

Detection_Date	Date	NOT NULL	The specific date the violation was discovered.
Statute_Deadline	Date	Calculated: Detection + 90d	The 3-month statutory deadline to institute proceedings.
Legal_Basis	String	ENUM ('Act', 'Regs', 'Both')	Maps directly to the "Prosecution Action" workflow decision node.
Violated_Codes	JSON Array	Mapped to Master Statute DB	Specific penal sections
Accused_Entity	String	NOT NULL	Name and address of the offending party.
Jurisdiction_Court	String	NOT NULL	The applicable Magistrate's Court.
Next_Calling_Date	Date	Nullable	Tracked date for the next judicial hearing.
Judicial_Verdict	String	ENUM ('Pending', 'Convicted', 'Discharged')	Final outcome of the Magistrate's ruling.
Imposed_Fine_LKR	Decimal	>= 0	Monetary penalty levied upon conviction.

Entity: Seizure_Destruction

This entity governs actions taken under the Food Act, specifically tracking the presence or absence of consent.

Field Name	Data Type	Constraint / Validation	Description
Seizure_ID	UUID	Primary Key, Auto-gen	Identifier for the seizure event.

Officer_ID	String	Foreign Key -> tbl_Officers	The Authorized Officer executing the seizure.
Inventory_Detail	Text	NOT NULL	Detailed manifest of the seized goods.
Seizure_Reason	String	NOT NULL	Justification (e.g., insect-infested, putrid, unregistered).
Owner_Consent	Boolean	NOT NULL	Flag indicating if written consent for destruction was obtained.
Consent_Signature	Blob	Nullable	Scanned document of owner consent.
Magistrate_Order	Blob	Required if Owner_Consent=FALSE	Scanned judicial order permitting destruction against owner's wishes.
Disposal_Date	Date	Nullable	Timestamp of physical destruction.

7. Import Control Module

The Import Control Module represents the fifth major functional component of the overarching the system architecture. This module is inextricably linked with the Sri Lanka National Single Window (NSW) and the Sri Lanka Customs Automated System for Customs Data (ASYCUDA) World platform depending on the technical feasibilities. The module is designed to digitize, streamline, and enforce the regulatory mechanism adopted by the FCAU at all national entry points. It facilitates coordination among multiple stakeholders. The Import Control Module addresses the end-to-end lifecycle of food importation, including importer registration, food classification, pre-arrival entries, document verification, border inspection, complex label compliance pathways, anonymized laboratory routing, and the processing of appeals for border rejections.

7.1 Comprehensive Workflow Elaboration

The Import Control Module digitizes a comprehensive, multi-staged workflow that involves preemptive checks, real-time border assessments, complex conditional logic for non-compliance, and post-inspection processing. This major workflow synthesizes the end-to-end procedural diagrams and operates in tandem with the Sri Lanka Customs ASYCUDA. The narrative below details the chronological sequence of events, actors, and system actions that constitute the primary operational pipeline.

Phase 01: Importer Registration and Pre-Import Entry

The workflow commences with **Importer Registration**. An importer must log into the portal and provide their Taxpayer Identification Number (TIN) or National Identity Card (NIC), alongside the precise geographical location (Geo-coordinates) of their primary storage warehouse. This geo-location data is critical for subsequent tracking if goods are released under conditional bonds. The importer uploads requisite documentation, such as the Business Registration Certificate, VAT certificate, and Details of the Director Board.

Following successful registration, the importer initiates the **Food Classification** process, selecting the specific Food Category and Subcategory for the intended consignment. This classification dictates the regulatory burden of the shipment. Subsequently, the importer files a **Pre-Import Entry** before the physical arrival of the goods. Upon submission, the system automatically generates a unique tracking identifier, the **FCAU No (e.g., I-XXXXXX)**.

Phase 02: Document Verification and Cross-Agency Integration

Triggered by the generation of the FCAU No, the system dynamically displays the **Required Documents & Regulations** specific to the selected food classification (e.g., Phytosanitary certificates, Certificates of Analysis, or SLS conformity documents). The importer proceeds to **Upload Documents** into the system.

The FCAU officers then perform Document Verification. This introduces the first major conditional node: **Approved?**

- **If No:** The system routes back to the importer with a Request Corrections notification, detailing the deficiencies. The importer must loop back to the document upload phase.
- **If Yes:** The workflow proceeds to the integration phase. The system triggers seamless Integration with Sri Lanka Customs (via ASYCUDA), the National Single Window (for payments) depending on the API availability.

With the electronic approvals synchronized, the Goods Arrive at Location (the physical port or airport).

Phase 03: Border Arrival, Registration, and Inspection

Upon the vessel's arrival and the subsequent lodging of the cargo manifest via ASYHUB, Customs Generates the Subject and Document Bundle and pushes this data via API to the FCAU module. The importer is then prompted for Registration Payment. The system interfaces with the government payment gateway to verify the fee and acquire Pay Approval. Once the payment is verified, the system executes the Register Entry command, formally logging the consignment into the active border queue.

This transitions the workflow to the physical Inspection Process conducted by Food and Drug Inspectors (FDIs) utilizing the H-800 form framework or specific import checklists to assess the cargo.

Phase 04: The Triage Decision: Release, Labeling, Sampling, or Destruction

Following the physical inspection, the FDI reaches a critical multi-path decision node: **Release / Sampling / Label Verification / Destroy**.

Path A: Label Verification (Compliance and Concessions)

If the FDI selects label verification, the system moves to **Label Verification (Compliance)**, checking against the Trilingual Regulations. This leads to a secondary condition node: **Supplementary Label / Reject / Release**.

- **Release:** If the labels are perfectly compliant, the workflow bypasses further checks and routes directly to High-Level Officer Approval.
- **Reject:** If the labeling is fundamentally flawed or fraudulent, the goods are rejected. The system notes that *Customs can recall after FCAU rejects, and the process is handled by customs*. The workflow terminates at the End node.
- **Supplementary Label:** If the non-compliance is curable (e.g., missing Sinhala/Tamil stickers on bulk imports), the system offers a concession. The goods are conditionally Released to warehouse, cannot re-sale. A default 1-month period is granted to the importer to affix supplementary labels, though the system allows FCAU to customize this time period. Once the importer completes the labeling, a Supplementary label verification is triggered, which loops the workflow back into the Inspection Process for a final check.

Path B: Sampling and Anonymized Laboratory Testing

If the physical inspection or the food's risk profile mandates testing, the workflow branches to the sampling protocol. The inspector performs a QR Scan Verification on the consignment to link the physical sample to the digital record. The system triggers Sampling, Investigate and other actions.

The module then prompts to Send Samples to Labs. Crucially, to prevent corruption and bias, the system handles the Payment for sampling and testing via the portal, but *the lab info is not visible to the Importer*. The system programmatically routes the sample to a public or private Chemical/Microbiological Laboratory.

The laboratory personnel log into their secure portal to Receive Samples to Lab, proceed to Conduct Testing, and finally Generate and Send Lab Report back to the FCAU module. Based on the analytical results, a condition node dictates the next step: Release / Re-test / Destroy.

- **Re-test:** Loops the process back to Send Samples to Labs.
- **Destroy:** Routes directly to the Enforcement/Follow-up phase and ends the process.
- **Release:** Routes to High-Level Officer Approval.

Path C: Immediate Destruction

If the goods are manifestly spoilt, insect-infested, or hazardous upon initial inspection, the FDI can route the workflow straight to **Destroy**, bypassing all testing and labeling loops, culminating in Enforcement/Follow-up.

Phase 05: Final Determination and Enforcement

Whether originating from a direct release, successful label verification, or passed laboratory tests, the consignment data flows to High-Level Officer Approval. The Chief Food Authority or a designated senior delegate reviews the digital dossier and authorizes the **Final Release**.

For consignments released under bond, or those requiring post-market surveillance, the system initiates an **Enforcement / Follow-up** protocol. This tracks the geo-location of the warehouse captured during registration to ensure that conditional goods (like those awaiting supplementary labels) have not entered the retail market prematurely. Upon completion of all regulatory actions, the major workflow reaches the **End** state.

7.2 Use Case Modeling and Narratives

The following use case models define the functional requirements and system behaviors necessary to support the complex regulatory workflows of the FCAU Import Control Module.

Use Case 01: Importer Registration and Geo-Mapping

Description: Allows an importer to register their business entity with the FCAU, cross-verifying TIN and VAT credentials, to obtain authorization for lodging food import declarations and establishing bonded warehouse geo-locations.

Primary Actor: Commercial Importer / Customs House Agent (CHA)

Secondary Actor: FCAU Registration Officer, Sri Lanka Customs System (ASYCUDA)

Pre-conditions: The Importer must hold a valid TIN and VAT Registration from the Department of Inland Revenue.

Post-conditions: An active Importer Profile is created in the FCAU module, the warehouse location is mapped, and the profile is synchronized with the ASYCUDA World registry (depending on the API availability).

Step	Action Flow
1	Importer accesses the FCAU portal and selects "New Importer Registration."
2	Importer enters TIN, VAT Number, NIC, and the geographical coordinates (Geo) of their primary storage warehouse.
3	Importer uploads mandatory documentation: Details of Director Board, Business Registration Certificate, and EDB certificates.
4	Importer submits primary contact details, including mobile numbers and email addresses for system notifications.
5	FCAU receives the application in the dashboard, reviews the attached documents, and executes the approval command.
6	System generates a unique FCAU Importer ID, updates the status to "Active," maps the warehouse geo-location, and transmits an approval payload to the ASYCUDA RAMS.

Use Case 02: Pre-Import Entry and Document Verification

Description: Manages the submission of cargo details prior to arrival, automates the determination of required regulatory documents based on food classification, and facilitates the initial FCAU document vetting process.

Primary Actor: Registered Importer

Secondary Actor: FCAU Document Verification Officer

Pre-conditions: The Importer must have an active profile (Use Case 01).

Post-conditions: The Pre-Import Entry is approved, generating an FCAU Number (I-XXXXXX), and the system transmits clearance readiness to ASYCUDA and Single Window.

Step	Action Flow
1	Importer initiates a "Pre-Import Entry" and selects the specific Food Category and Subcategory.
2	System generates a unique FCAU No (e.g., I-XXXXXX) for the consignment workflow.
3	System dynamically displays the specific "Required Documents & Regulations" checklist based on the selected classification.
4	Importer uploads the requested documents (e.g., Commercial Invoice, Certificates of Analysis, Phytosanitary certificates).
5	FCAU Officer accesses the "Document Verification" queue and reviews the uploaded artifacts.
6	FCAU Officer evaluates the entry. <i>Decision:</i> Approved? (Yes/No).
7a	System (If no) Triggers "Request Corrections," sending an automated email to the importer detailing deficiencies. Workflow loops back to Step 4.
7b	System (If yes) Updates status to "Approved" and executes "Integration with Customs (ASYCUDA) and Single Window (Payments)."

Use Case 03: Border Inspection and Label Verification

Description: The process by which border inspectors evaluate incoming consignments, verify physical packaging against Trilingual regulations, and manage the supplementary labeling concession workflow.

Primary Actor: Food and Drug Inspector (FDI)

Secondary Actor: Importer, ASYCUDA System

Pre-conditions: Goods have physically arrived at the location, registration payments are complete, and the "Register Entry" node has been passed.

Post-conditions: The consignment is released for clearance, rejected entirely, or released conditionally to a warehouse for supplementary labeling.

Step	Action Flow
1	FDI accesses the "Inspection Process" module at the border entry point.
2	FDI selects "Label Verification" path to assess compliance with the Food (Labelling and Advertising) Regulations.
3	FDI inputs findings into the system. <i>Decision:</i> Is the label fully compliant, resolvable, or fundamentally flawed?
4a	System (If Compliant) routes the workflow directly to "High-Level Officer Approval" for final release.
4b	System (If Flawed) routes to "Reject." Notifies Customs for cargo recall and updates status to terminate the import.
4c	FDI (If Resolvable) Selects "Supplementary Label" concession.
5	System prompts the FDI to set a compliance deadline (Default is 1 month, but customizable).
6	System generates a conditional release order: "Released to warehouse, cannot re-sale." System flags the warehouse geo-coordinates for follow-up.
7	Importer affixes supplementary labels at the warehouse and requests re-inspection via the portal.
8	FDI conducts "Supplementary label verification." If passed, routes back to the main Inspection Process release node.

Use Case 04: Anonymized Sampling and Laboratory Analysis

Description: Orchestrates the drawing of physical samples, secure financial transactions for testing fees, and the blind routing of samples to laboratories to ensure impartial analytical results.

Primary Actor: Food and Drug Inspector (FDI), Chemical/Microbiological Laboratory

Secondary Actor: Importer

Pre-conditions: The inspection process dictates that chemical or microbiological testing is required.

Post-conditions: A lab report is generated, leading to the release, destruction, or re-testing of the goods.

Step	Action Flow
1	FDI selects the "Sampling" path during the Inspection Process.
2	FDI scans the consignment utilizing the "QR Scan Verification" tool to link the physical sample to the digital record.
3	System generates a payment request for "Sampling and Testing Fee" to the Importer.
4	Importer completes the payment via the Single Window payment gateway.
5	System automatically routes the sample dispatch order to an available lab. <i>Crucial:</i> The system obscures the laboratory's identity from the Importer to prevent bias/tampering.
6	Laboratory logs into the lab portal, clicks "Receive Samples to Lab," and updates custody status.
7	Laboratory conducts analytical testing and utilizes the "Generate and Send Lab Report" function to upload results (e.g., Aflatoxin, Heavy Metals).
8	System evaluates the lab report. <i>Decision:</i> Release / Re-test / Destroy.
9	System routes the decision accordingly (Release to High-Level Approval, Re-test loops to Step 5, Destroy loops to Enforcement).

Use Case 05: High-Level Approval and Post-Market Enforcement

Description: The final authorization sequence where senior FCAU officials review the aggregated data of a consignment before permitting entry into the Sri Lankan market, followed by post-clearance enforcement tracking.

Primary Actor: High-Level Officer (e.g., Chief Food Authority / DDG)

Secondary Actor: Enforcement Officers

Pre-conditions: A consignment has successfully navigated the Label Verification or Laboratory Testing workflows and is queued for final release.

Post-conditions: The ASYCUDA is updated to 'Green' (Cleared), and post-market surveillance tasks are generated if necessary.

Step	Action Flow
1	High-Level officer Accesses the "High-Level Officer Approval" dashboard.
2	System aggregates and displays the Pre-Import Entry, Document Verification logs, Inspection outcomes, and Lab Reports in a unified dossier.
3	High-Level Officer reviews the dossier and executes the "Final Release" command.
4	System transmits an API payload to ASYCUDA RAMS, updating the selectivity color code to 'Green', allowing Customs to release the goods from the port.
5	System checks if the consignment requires post-market tracking (e.g., goods previously under a Supplementary Label bond).
6	System if tracking is required, generates tasks in the "Enforcement / Follow-up" queue for regional Public Health Inspectors (PHIs).
7	Enforcement Officer conducts field visits based on the generated tasks and closes the loop in the system. Workflow ends.

7.3 Data Dictionary and Entity-Relationship Architecture

The following data dictionaries define the core entities required for operations, ensuring seamless mapping to ASYCUDA XML payloads, Single Window interactions, and laboratory reporting schemas.

Entity: Importer_Profile

Stores the verified registration details of commercial food importers and their spatial data.

Field Name	Data Type	Constraints	Description
Importer_ID	VARCHAR	Primary Key	Unique system generated identifier for the importer.
TIN_Number	VARCHAR	Not Null, Unique	Taxpayer Identification Number issued by Inland Revenue.
VAT_Number	VARCHAR	Not Null	Value Added Tax registration number.
NIC_Number	VARCHAR	Nullable	National Identity Card number for Sole Proprietorships.
Company_Name	VARCHAR	Not Null	Legal name of the business entity.
Warehouse_GeoLat	DECIMAL	Not Null	Latitude coordinate of the primary storage location.
Warehouse_GeoLon	DECIMAL	Not Null	Longitude coordinate of the primary storage location.
BR_Cert_No	VARCHAR	Not Null	Business Registration Certificate Number.
Contact_Person	VARCHAR	Not Null	Designated representative for customs queries.
Mobile_SMS	VARCHAR	Not Null	Verified mobile number.
Email_Alert	VARCHAR	Not Null	Official email address for documentation routing.

Profile_Status	ENUM	Default 'Pending'	Values: 'Pending', 'Active', 'Suspended', 'Revoked'.
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Entity: Import_Consignment_Entry

Captures the data required for evaluating commodities from Pre-Import Entry through to Final Release.

Field Name	Data Type	Constraints	Description
FCAU_No	VARCHAR	Primary Key	Unique identifier (e.g., I-XXXXXX) generated at Pre-Import.
Importer_ID	VARCHAR	Foreign Key	Links to Importer_Profile.
CUSDEC_No	VARCHAR	Nullable, Index	Customs Declaration Number synced from ASYCUDA upon arrival.
Food_Category	VARCHAR	Not Null	Primary classification (e.g., Dairy, Spices, Confectionary).
Food_Subcategory	VARCHAR	Not Null	Secondary classification for precise rule routing.
HS_Code	VARCHAR	Not Null	Harmonized System Code used for risk profiling.
Doc_Verif_Status	ENUM	Default 'Pending'	Values: 'Pending', 'Request_Corrections', 'Approved'.
Payment_Status	ENUM	Default 'Unpaid'	Status of the Registration/Inspection Fee (Gov Pay Approval).
Workflow_Stage	VARCHAR	Not Null	Current node (e.g., 'Inspection', 'Lab_Testing', 'Final_Release').

Entity: Border_Inspection_Log

Logs the real-time operational data captured by FDIs during the physical evaluation of goods.

Field Name	Data Type	Constraints	Description
Inspection_ID	VARCHAR	Primary Key	Unique identifier for the inspection event.
FCAU_No	VARCHAR	Foreign Key	Links to Import_Consignment_Entry.
FDI_Officer_ID	VARCHAR	Not Null	System ID of the inspecting Food and Drug Inspector.
Trilingual_Label	BOOLEAN	Not Null	TRUE if label possesses English, Sinhala, and Tamil text.
Date_Of_Manufacture	DATE	Not Null	Extracted from physical packaging.
Date_Of_Expiry	DATE	Not Null	Extracted from physical packaging.
Shelf_Life_Pct	DECIMAL	Computed	Automated calculation: (Expiry - Current Date) / (Expiry - DOM). Must be >= 60.00.
Label_Decision	ENUM	Nullable	Values: 'Compliant', 'Reject', 'Supplementary_Label'.
Supp_Label_Deadline	DATE	Nullable	Deadline for corrective labeling (Default 1 month from inspection).
Inspection_Route	ENUM	Not Null	Next workflow step: 'Release', 'Sampling', 'Destroy'.

Entity: Laboratory_Analysis_Record

Manages the anonymized routing and resulting data points of physical samples sent for testing.

Field Name	Data Type	Constraints	Description
Sample_ID	VARCHAR	Primary Key	Unique tracking barcode generated during QR Scan Verification.
FCAU_No	VARCHAR	Foreign Key	Links to Import_Consignment_Entry.
Assigned_Lab_ID	VARCHAR	Not Null	Internal ID of the assigned lab (hidden from importer UI).
Test_Type	VARCHAR	Not Null	e.g., 'Microbiological', 'Chemical', 'Aflatoxin'.
Testing_Fee_Paid	BOOLEAN	Default FALSE	Ensures the importer has paid prior to lab dispatch.
Date_Received	DATETIME	Nullable	Timestamp when the lab formally receives the physical sample.
Report_URL	VARCHAR	Nullable	Encrypted cloud link to the generated Lab Report PDF.
Lab_Result	ENUM	Nullable	Values: 'Pass', 'Fail', 'Inconclusive'.
System_Action	ENUM	Nullable	Workflow trigger based on result: 'Release', 'Re-test', 'Destroy'.

8. Export Control Module

The Export Control Module is engineered to govern the entire lifecycle of food export compliance. Its scope encompasses the initial statutory registration of food manufacturing and exporting premises, the rigorous processing of individual export health certificate applications, the orchestration of physical sampling, the integration of microbiological and chemical laboratory testing, and the final cryptographic issuance of legally binding health certificates.

8.1 Comprehensive Workflow Elaboration

The core business logic of the Export Control Module is structured around a sequential, multi-actor procedural flow. The system architecture coordinates the interactions of three primary actors: the Exporter, the FCAU (acting as the regulatory gatekeeper and auditor), and the Chemical / Microbiological Laboratory (acting as the independent scientific verification node). The following elaboration details the precise progression of states, decision nodes, and data handoffs as defined by the operational flow diagram and statutory guidelines.

Phase 1: Business Registration and Credentialing

The export compliance lifecycle strictly mandates that an entity cannot apply for a consignment-specific health certificate without first possessing a validated, registered manufacturing or warehousing establishment. The workflow commences with the Exporter actor initiating the "Register the Business with FCAU" process.

The system presents a comprehensive digital interface for the submission of mandatory corporate and operational data. The Exporter must "Submit Required Documents," which the system securely ingests and indexes. The mandatory digital artifacts include the Certificate of Incorporation (Business Registration), the Taxpayer Identification Number (TIN) Certificate, the Value Added Tax (VAT) Certificate, and specific declarations detailing the corporate directorate. To ensure data integrity, the system utilizes API integrations within the National Single Window to cross-verify the TIN and VAT credentials directly against the Inland Revenue Department's active databases.

Beyond basic fiscal identity, the module necessitates the submission of ISO and Quality Certificates. Depending on the risk profile of the food items manufactured, the exporter uploads proof of Good Manufacturing Practices (GMP), Hazard Analysis and Critical Control Points (HACCP) system implementation, ISO 22000 certifications, and any commodity-specific registrations.

Once the information is submitted, the workflow transitions to the FCAU actor for the "FCAU Reviews Documents" phase. Authorized Food and Drug Inspectors access a dedicated dashboard to evaluate the submitted credentials. If the documentation satisfies preliminary regulatory requirements, the system schedules a "Physical Factory Inspection." This is a critical offline process managed digitally by the module. Inspectors utilize a mobile-responsive assessment tool to conduct the audit, scoring the facility across seven stringent domains defined by the FCAU: Building and Surroundings; Hygiene, Biosecurity, and Employee Training; Sanitation and Pest Control; Process Controls; Equipment Design; Transportation and Storage; and Traceability.

The inspector inputs the categorical scores into the system. If the cumulative assessment meets the prescribed hygiene thresholds, the system's state machine progresses to the "Factory Registration Number Issued" node. The module automatically generates a unique, time-bound registration identifier (e.g., FCAU-EXP-2026-XXXX), permanently associating it with the exporter's digital profile, thereby unlocking the entity's authorization to request specific consignment certificates.

Phase 2: Export Certificate Application and Financial Settlement

With a validated establishment profile, the Exporter can initiate the transactional phase: "Apply for Export Certificate (Submit Application Form)." This phase represents the core of the module's daily operations. The exporter inputs granular data regarding the specific consignment intended for international transport.

The application form fields require meticulous detail to ensure full supply chain transparency. The exporter must declare the total quantity, net weight, exact nature of the food, lot identifiers, packaging types, and the specific destination country. Crucially, the application requires the attachment of raw material sourcing declarations, existing product analysis certificates, and proforma invoices. For specific commodities, third-party attestations must be uploaded, such as a National Organic Control Unit (NOCU) certificate for organic items.

Concurrently with or immediately following the application submission, the workflow encounters the "Payment handling" node. The system calculates the statutory processing, inspection, and certification fees based on the declared FOB (Free On Board) value of the consignment or a fixed schedule of charges. To facilitate seamless trade, the Export Control Module integrates directly with the Online Payment Platform. The system generates a digital payment mandate (incorporating an IECD/FCAU Reference Number) and routes the exporter to their preferred digital banking unit. The transaction is settled in real-time via the Common

Electronic Fund Transfer Switch (CEFTS), and a cryptographic success token is instantly returned to the module, clearing the application for technical review.

Phase 3: The Sampling Decision Matrix and Laboratory Handover

Upon successful payment and application submission, the workflow reaches a critical internal decision node executed by the FCAU: "Is Sampling Required?".

If the reviewing FCAU officer determine that sampling is *not* required (the "No" path), the workflow immediately bypasses laboratory integration and proceeds directly to the final "If Compliant" review node.

However, if the determination is "Yes," the workflow dictates that a "Sample Collected" event must occur. An FCAU field officer is dispatched to the warehousing facility to draw a representative physical sample from the consignment lot. The officer utilizes the mobile responsive module to log the physical sample, generating a unique QR code that is physically affixed to the sample container. This ensures blind testing and absolute chain-of-custody traceability.

The sample is then "Sent to Laboratory." The Export Control Module includes a dedicated, secure interface for the third actor: the "Chemical / Microbiological Laboratory (Public / Private)." The system recognizes an ecosystem of authorized laboratories capable of testing formal samples under the Food Act.

The laboratory workflow begins with "Receive Samples to Lab." The lab technician scans the QR code, The laboratory then proceeds to "Conduct Testing" in the physical world. Upon completion, the technician utilizes the portal to "Generate and Send Lab Report." The qualitative and quantitative results, alongside the formalized PDF analysis report, are uploaded directly into the module's database, permanently attaching them to the specific consignment record.

Phase 4: Compliance Verification and Resolution Loops

The return of the laboratory data triggers the pivotal "If Compliant" decision node within the FCAU application logic.

The flow diagram defines three distinct outcomes from this compliance node:

1. **If Compliant -> Yes:** The laboratory results fall within all acceptable parameters, and the documentation is flawless. The system automatically advances the application to the issuance phase.

2. **If Compliant -> No:** The laboratory detects critical contamination. The consignment poses a severe health risk or violates international trade laws. The workflow routes to "End," resulting in an outright rejection of the application. The system generates a formal rejection notice, alerts the exporter, and may automatically flag the establishment for an emergency compliance audit.
3. **Corrections Required:** The flow diagram indicates a specific feedback loop where the application is not immediately rejected but requires remediation. This arrow routes back toward the sampling or documentation phases. This occurs when minor administrative discrepancies are found (e.g., mismatched lot numbers on the invoice versus the physical product) or when a lab result is marginal and the exporter exercises their legal right to request a secondary test from a different authorized laboratory. The system places the application in a "Suspended - Pending Correction" state, automatically detailing the specific defect to the exporter and awaiting an updated submission or a re-sampling directive.

Phase 5: Certificate Issuance and Trade Execution

For applications that successfully navigate the compliance checks (the "Yes" pathway), the system enters the final state: "Health Certificate Issued (Within 3-4 days)." This strict Service Level Agreement (SLA) is critical for perishable commodities and overall trade velocity.

The Export Control Module dynamically compiles the approved data into the standardized international format for an Export Health Certificate. The document includes the precise attestation clauses required by the importing nation, the lot identifiers, the FCAU official stamp, and the certifying officer's details.

The system issues the finalized, signed scanned PDF to the exporter's portal. Furthermore, acting as a node within the National Single Window, the module transmits the certificate's structured metadata directly to Sri Lanka Customs' ASYCUDA World system. This backend integration allows customs agents to instantly verify the SPS compliance of the cargo at the Export Facilitation Centre or Air Cargo terminal without relying on easily forged paper documents.

With the digital clearance secured, the workflow reaches the final operational step: "EXPORT FOOD ITEMS," culminating in the successful cross-border dispatch of the consignment, concluding the module's workflow at the "End" node.

8.2 Use Case Modeling and Narratives

To translate the overarching architectural workflow into precise, actionable software requirements, the system is modeled through a series of exhaustive use cases. These narratives define the exact interactions between the human actors, the external system APIs, and the Export Control Module's internal logic engine.

Use Case 1: Exporter Establishment Registration and Credentialing

Description: The comprehensive procedure by which a commercial entity submits corporate identity, fiscal credentials, and facility quality standards to the FCAU, undergoes physical auditing, and successfully obtains a unique Factory Registration Number, authorizing them to operate as a certified food exporter.

Primary Actor: Exporter (Authorized Corporate Representative).

Secondary Actors: FCAU Food and Drug Inspector, IRD API (Inland Revenue Department).

Pre-conditions: The Exporter is a legally incorporated entity in Sri Lanka with valid TIN and VAT registrations. The Exporter has created a basic user profile within the National Single Window portal.

Post-conditions: The Exporter profile is updated with a valid Est_Reg_Number. The facility is marked as compliant, and the user interface unlocks the capability to initiate Health Certificate applications.

Step	Action / Workflow
1	Exporter selects "Register the Business with FCAU" from the main dashboard. Inputs primary corporate metadata: Company Name, Business Registration Number, and Date of Incorporation. System validates string formats and queries the internal database to prevent duplicate entity registrations.
2	Exporter inputs the Taxpayer Identification Number (TIN) and Value Added Tax (VAT) certificate numbers.

	Initiates a synchronous API call to the IRD database. Validates the active status of the fiscal IDs and returns a confirmation token.
3	Exporter enters precise geographical coordinates and physical address text for the Corporate Office, the Manufacturing Factory, and the Storage Warehouse.
4	Exporter uploads mandatory digital artifacts: Director Details, Business Registration PDF, and specific commodity board registrations (e.g., Sri Lanka Tea Board, Coconut Development Authority).
5	Exporter submits "ISO / Quality Certificates" data, detailing active GMP, HACCP, or ISO 22000 implementations, including expiry dates and scanned proofs.
6	Exporter reviews the final application summary, accepts the legal declaration of truthfulness, inputs the verification captcha, and clicks "Submit".
	System generates a unique Application Tracking ID. Dispatches a confirmation email to the Exporter. Transitions the application state to Pending_FCAU_Review and routes it to the Inspector's queue.
7	FCAU Inspector accesses the dashboard, opens the specific application, and executes "FCAU Reviews Documents." System renders all submitted data and PDF viewers side-by-side for verification.
8	FCAU Inspector Approves the documentation phase and clicks "Schedule Physical Factory Inspection." System generates an automated notification to the Exporter detailing the inspection date and required on-site preparations. Changes state to Pending_Audit.
9	FCAU Inspector conducts the physical site visit. Utilizes the module's mobile interface to input qualitative and quantitative scores across the seven hygiene and process control domains. System captures real-time data inputs. Calculates the aggregate compliance score based on weighted algorithms.
10	FCAU Inspector submits the final audit report and selects "Approve Establishment." System updates the database. Generates the official "Factory Registration Number."
11	System Executes the final registration protocol.

	Issues a signed Certificate of Registration to the Exporter. Updates the module access control list, unlocking Phase 2 features for the user.
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Use Case 2: Health Certificate Application and Financial Settlement

Description: The process whereby a registered exporter submits detailed consignment data, required documentation, and completes the statutory fee payment to request an Export Health Certificate for a specific international shipment.

Primary Actor: Registered Exporter.

Secondary Actors: Online Payment Platform

Pre-conditions: The Exporter possesses an active, unexpired Factory Registration Number. The intended export commodity falls under the legal jurisdiction of the FCAU.

Post-conditions: The application is formally lodged, the financial transaction is irrecoverably settled, and the information is queued for the FCAU sampling determination matrix.

Step	Action / Workflow
1	Exporter Clicks "Apply for Export Certificate" and selects the specific registered facility from which the goods originate. System Validates the active status of the facility. Renders the dynamic application form.
2	Exporter Selects the "Destination Country" from an ISO-standardized dropdown and defines the specific "Nature of the Food" and "Intended Purpose". Queries the rules engine to load specific SPS constraints, required attestations, and mandatory lab testing parameters dictated by the selected destination.
3	Exporter Inputs exact logistical and quantitative data: Total Quantity, Number of Packages, Net Weight, Type of Packaging, and precise Lot Identifiers. System Performs data typing validation (e.g., ensuring Net Weight is a rational decimal).
4	Exporter Uploads supporting transactional documents: Proforma Invoice, Country of Origin certificate, existing Product Analysis Certificates, and organic/commodity specific declarations.

5	Exporter Submits the completed application form. System Locks the form data. Transitions state to Pending_Payment. Calculates the required inspection/certification fee based on the fee schedule. Generates the IECD/FCAU Piv No and DocRef No.
6	Exporter Reviews the fee calculation and clicks the "Payment handling" execution button. Redirects the user's session securely to the Online Pay API gateway, passing the transaction metadata and amount.
7	Exporter Interacts with their selected digital banking interface and confirms the fund transfer. (External Process within Bank/IPG infrastructure).
8	IPG (External) Processes the transaction via the CEFTS network. Transmits a secure, encrypted webhook back to the Export Control Module containing the transaction success status, timestamp, and Cyber Receipt Number.
9	System Receives the IPG success payload. Updates the financial ledger. Attaches the receipt to the application. Transitions the application state to Pending_FCAU_Review.

Use Case 3: FCAU Application Review and Sampling Determination

Description: The internal processing by the competent authority to review the submitted consignment data and decide whether physical sampling and laboratory testing are mandatory before certificate issuance.

Primary Actor: FCAU Evaluating Officer, FCAU Field Inspector.

Pre-conditions: A fully paid, complete application has been submitted by the Exporter.

Post-conditions: The application is either routed directly to final compliance checks or paused while a physical sample is drawn and dispatched to an authorized laboratory.

Step	Action / Workflow
1	FCAU Officer Accesses the application queue and opens the "FCAU Reviews Application" interface for a newly submitted, paid dossier. System Renders all exporter inputs, attached invoices, and historical compliance data for the specific factory.
2	FCAU Officer Reviews the documentation against the destination country's SPS requirements. Initiates the "Is Sampling Required?" decision matrix.
3	FCAU Officer Selects "Yes - Sampling Required." Changes application state to Pending_Sampling. Generates a physical sampling task order and assigns it to a Field Inspector based on the factory's geographical location.
4	FCAU Inspector Travels to the warehouse, locates the specified Lot Identifier, and executes the physical sampling protocol. (N/A (Physical action)).
5	FCAU Inspector accesses the specific task order, and clicks "Sample Collected." Generates a unique, secure QR code for the sample
6	FCAU Inspector Prints and affixes the QR code label to the sample container, and clicks "Sent to Laboratory," selecting the specific authorized testing facility. Dispatches an electronic notification to the selected Laboratory Portal. Changes application state to Awaiting_Lab_Results.

(Note: If Step 3 results in a "No" determination, the workflow jumps directly to Use Case 5, Step 1).

Use Case 4: Laboratory Sample Reception, Testing, and Reporting

Description: The coordination of public or private authorized laboratories into the workflow. The lab receives the blinded sample, conducts the necessary chemical and microbiological tests, and feeds the scientific data back into the module's compliance engine.

Primary Actor: Laboratory Technician (Public / Private Authorized Lab).

Secondary Actors: FCAU System Engine.

Pre-conditions: A physical sample has been logged by the FCAU and delivered to the laboratory facility. The application is in the Awaiting_Lab_Results state.

Post-conditions: Comprehensive analytical data is structured and saved within the system.

Step	Action / Workflow
1	Lab Analyst accesses the "Receive Samples to Lab" interface. Scans the physical QR code affixed to the sample. Acknowledges receipt of the sample in the system.
2	Lab Analyst Conducts the physical "Conduct Testing" phase utilizing laboratory equipment. N/A (External physical process).
3	Lab Analyst Returns to the portal to execute "Generate and Send Lab Report." Uploads the formalized, signed PDF analysis report
4	Lab Analyst Reviews the data entry and clicks "Submit Results to FCAU." Locks the laboratory record from further editing. Routes the structured data payload back to the core module's compliance engine.

Use Case 5: Compliance Verification, Loop Resolution, and Certificate Issuance

Description: It manages the "If Compliant" decision logic, handles the "Corrections Required" feedback loops, and ultimately issues the cryptographically secured Health Certificate.

Primary Actor: FCAU Evaluating Officer

Secondary Actors: ASYCUDA World System (Sri Lanka Customs).

Pre-conditions: All required documentation is present, payment is settled, and (if mandated) laboratory results have been submitted.

Post-conditions: The final "Health Certificate" is issued, enabling the Exporter to clear customs, or the application is definitively rejected.

Step	Action / Workflow
1	System Receives the lab results
2	FCAU Officer Accesses the application for final review.
3	FCAU Officer Identifies a minor discrepancy or an anomalous lab result that requires re-verification. Selects the "Corrections Required" pathway. Changes status to Suspended. Opens a communication module to detail the required remediation to the Exporter. Generates a "Re-Sample" order, routing the workflow back to Use Case 3, Step 4.
4	(Alternative to Step 3) FCAU Officer Identifies a critical failure (e.g., severe contamination). Selects the "If Compliant -> No" pathway. Changes status to Rejected. Generates a formal rejection letter. Terminates the workflow at the "End" node.
5	(Alternative to Step 3 & 4) FCAU Officer Verifies that all metrics are perfect. Selects the "If Compliant -> Yes" pathway and clicks "Authorize Issuance." Triggers the final certificate generation microservice.
6	System Executes the "Health Certificate Issued" protocol within the mandated 3-4 day SLA window. Compiles the PDF document featuring the FCAU official stamp and Certifying Officer details.
7	System Finalizes the distribution of the digital asset. Transmits the certificate securely to the Exporter's dashboard. Pushes an XML/JSON payload containing the certificate metadata directly to the Sri Lanka Customs ASYCUDA World system to facilitate immediate border pre-clearance. Transitions the workflow to the final "EXPORT FOOD ITEMS" / "End" state.

8.3 Data Dictionary and Entity-Relationship Architecture

The functional integrity and interoperability of the Export Control Module rely on a rigorously defined, highly structured data architecture. This architecture must accommodate the diverse

requirements of the FCAU, the financial tracking, laboratory reporting, and the supply chain standards of the National Single Window. The following Data Dictionary defines the specific fields, constraints, formats, and ownership parameters for the primary system entities.

Entity 1: exporter_establishment_profile

This entity serves as the foundational credential repository for any business attempting to export food. It manages the data required for the "Register Business with FCAU" and "Physical Factory Inspection" phases.

Field Name	Data Type / Constraints	Description
Est_Reg_Number	Varchar(30) (e.g., FCAU-EXP-2026-0012) Primary Key, Auto-generated, Immutable	The unique primary key generated by the system upon successful FCAU inspection.
Company_Name	Varchar(150) Mandatory	The full legal operating name of the manufacturing or exporting entity.
Biz_Reg_Number	Varchar(50) Mandatory, Unique	Official Business Registration or Certificate of Incorporation identifier
TIN_Number	Varchar(20) Mandatory, Verified via API	Taxpayer Identification Number used for cross-agency fiscal verification
VAT_Number	Varchar(20) Conditional, Verified via API	Value Added Tax registration number.
Date_Of_Incorporation	Date (YYYY-MM-DD) Mandatory	The official date the business was legally formed.

Nature_Of_Company	Enum	Mandatory	Classification of the business operation (e.g., "Manufacturer & Exporter", "Only Exporter", "Re-Export").
Corporate_Address	Text	Mandatory	The registered physical address of the company headquarters.
Factory_Address	Text	Mandatory	The precise physical address of the manufacturing facility subjected to FCAU inspection.
Premises_Reg_No	Varchar(50)	Mandatory	Prior registration identifier obtained under the Food (Registration of Premises) Regulations
Director_Details	JSON	Mandatory	A structured array containing the names, titles, direct phone numbers, and emails of the Heads of Institute
Commodity_Board_Regs	JSON	Conditional	Key-value pairs for specific statutory board registrations (e.g., {"SLTB": "T-1234", "CDA": "C-987"})
Quality_Standard_Certs	JSON Array	Mandatory	A structured array of active quality systems (e.g., GMP, ISO 22000, HACCP), including certificate numbers and expiry dates.
Production_List	Text	Mandatory	Categorized text or code list of all food items manufactured by the facility

Local_Market_Availability	Boolean	Mandatory	Boolean flag indicating if the manufactured product is also freely available in the domestic Sri Lankan market
Exporting_Countries	JSON Array	Mandatory	Array of ISO 3166-1 alpha-2 country codes representing the intended global markets for the entity's products
Establishment_Status	Enum	Mandatory	The current operational state of the profile (e.g., Pending_Audit, Active, Suspended, Revoked).

Entity 2: export_consignment_application

This entity models the transactional data generated when an exporter initiates the "Apply for Export Certificate" process. It captures the highly specific logistical and descriptive data required to satisfy international border control authorities.

Field Name	Data Type / Format	Constraints	Description
Application_ID	Varchar(40) (UUID)	Primary Key, Auto-generated	The unique tracking identifier for the specific certificate request.
Est_Reg_Number	Varchar(30)	Mandatory	Foreign key linking the application to the specific approved factory profile.
Product_Name	Varchar(200)	Mandatory	The exact commercial and technical description of the food product being exported.

Species_Code	Varchar(100)	Conditional	The specific scientific nomenclature (e.g., utilizing the FAO ASFIS database) required for specific commodities like dried marine products.
Intended_Purpose	Varchar(100)	Mandatory	Declaration of how the product will be used in the destination country (e.g., "Human Consumption", "Further Processing").
Lot_Identifier	Varchar(100)	Mandatory	The specific alphanumeric production batch or lot code crucial for international traceability and recalls.
Total_Quantity	Integer	Mandatory	The discrete integer count of items or primary packages in the consignment.
Net_Weight	Decimal(12,3)	Mandatory	The precise total weight of the edible product, excluding all packaging materials.
Packaging_Type	Varchar(150)	Mandatory	Text description of the containment method (e.g., "Vacuum sealed polyethylene pouches in corrugated cartons").
Destination_Country	Varchar(2)	Mandatory	The ISO 3166-1 alpha-2 code of the sovereign jurisdiction receiving the consignment.
QR_Code_ID	Varchar(100)	Optional	Unique secure QR code generated.
Proforma_Invoice_URL	URL	Mandatory	Secure system path to the uploaded commercial invoice document.

Raw_Material_Certs_URL	URL	Conditional	Secure system path to documents proving the origin and safety of raw materials used in processing.
Application_Status	Enum	Mandatory	The real-time state of the workflow (e.g., Draft, Pending_Payment, Pending_Sampling, Awaiting_Lab_Results, Compliant, Rejected).

Entity 3: laboratory_analysis_payment_transaction

This entity integrates the external processes crucial to the module's functioning: the scientific verification conducted by the "Chemical / Microbiological Laboratory" and the financial settlement processed via the IPG integration.

Field Name	Data Type / Constraints Format	Description
Record_ID	Varchar(40) (UUID) Primary Key	Unique identifier for this combined financial and scientific record.
Application_ID	Varchar(40) Mandatory	Foreign key linking back to the parent Consignment Application.
Payment_IECD_Piv_No	Varchar(30) Mandatory	The unique PIV number generated by the system and passed to IPG to identify the billing entity.
Payment_DocRef_No	Varchar(50) Mandatory	The document reference number utilized during the CEFTS routing process.

Payment_Amount	Decimal(10,2) (LKR)	Mandatory	The calculated statutory fee collected for processing and certification (e.g., calculated as 0.2% of FOB value).
Receipt_No	Varchar(100)	Mandatory	The cryptographic cyber receipt number returned by the Online Payment Platform upon successful fund realization.
Payment_Timestamp	DateTime	Mandatory	The precise server time the transaction was cleared through the CEFTS network.
Sample_QR_code	Varchar(100)	Conditional	The unique QR code tracker affixed to the physical sample by the FCAU inspector, used to blind the sample for the lab.
Lab_ID	Varchar(30)	Conditional	The unique identifier of the specific public or private authorized laboratory executing the tests.
Lab_Report_URL	URL	Conditional	The secure system path to the formalized PDF analysis report uploaded by the lab analyst.
System_Compliance_Flag	Boolean	Conditional	The binary output of the rules engine after comparing lab data against destination MRLs (Pass/Fail).

Entity 4: export_health_certificate

The final, legally binding output of the system, representing the culmination of the entire workflow. This entity holds the data that is digitally signed and transmitted to customs authorities.

Field Name	Data Type / Constraints Format	Description
Cert_ID	Varchar(50) Primary Key, Auto-generated	The officially recognized, unique alphanumeric identifier for the Health Certificate.
Application_ID	Varchar(40) Mandatory, Immutable	Foreign key linking to the full dossier of evidence that justified the certificate's issuance.
Issue_Date	Date (YYYY-MM-DD) Mandatory, Immutable	The date the FCAU officer authorized the certificate.
Expiry_Date	Date (YYYY-MM-DD) Mandatory, Immutable	The calculated date beyond which the certificate is considered legally invalid by importing authorities.
Certifying_Officer_Name	Varchar(150) Mandatory	The legal name of the FCAU official who executed the final approval.
Certifying_Officer_Title	Varchar(100) Mandatory	The official position of the authorizing officer.
Attestation_Clauses	Text Mandatory	The specific legal text appended to the certificate certifying that the product meets the importing nation's SPS requirements (e.g., "Fit for human consumption").
Lab_Report_URL	URL Conditional	The secure system path to the formalized PDF certificate uploaded by authorizing officer.

ASYCUDA_Sync_Status	Boolean	Mandatory	Boolean flag indicating whether the certificate metadata has been successfully pushed to the Sri Lanka Customs system.
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9. Registration of Common Salt Manufacturing and Transports Management Module

This provides an exhaustive, highly detailed architectural and functional blueprint for the salt manufacturing, Saltern registration, and transport permit ecosystem. The module is intricately engineered to enforce statutory compliance. The successful implementation of this system requires the orchestration of inter-agency workflows, seamlessly connecting the Sri Lanka Standards Institution, Medical Officers of Health, field-deployed Public Health Inspectors, laboratory analysts, and the national digital payment gateway.

9.1 Comprehensive Workflow Elaboration

The core functionality of the registration module is defined by five distinct, interrelated operational workflows: New Saltern Registrations, New Factory Registrations, Registration Renewals, Registration Amendments, and the issuance of Transport Permits. The system architecture coordinates these processes across a distributed, national network of actors, utilizing a strict state-machine model to govern application progression.

Phase 01: Saltern and Factory Registration Workflow

The initiation of a new registration for either a raw salt extraction Saltern or a processing factory requires the applicant to first interact with the public-facing portal. The workflow commences with the user executing the "Sign up to the system" function, followed by the "Create Profile" and "Login to the Profile" actions.

Once authenticated, the applicant initiates the core data entry phase, defined as "Fill Application and Upload required Documents". The software must support the ingestion of multiple product categories simultaneously and accommodate complex document payloads. Mandatory digital uploads include a certified copy of the business registration, the national identity card of the primary business owner, approved architectural building plans, local authority registration certificates, high-resolution product labels, explicit site plans marked on survey plans and detailed factory layouts.

Upon successful data submission, the application state transitions to the central authority, triggering the "Receive application" and subsequent "Review and Validate the application" nodes by the designated reviewing officer. If the preliminary digital dossier is compliant, the system generates a dynamic "Payment Request." A critical architectural requirement here is the algorithmic calculation of the fee structure; the system must compute a base registration fee

combined with a variable inspection fee that differs proportionally based on the geographical distance to the facility. Following successful payment realization via the integrated gateway, the system automatically routes the digital file to the respective district-level Food and Drugs Inspector to "Conduct site inspection".

The field inspection phase is entirely digitized. The inspector utilizes a mobile responsive interface to conduct the site visit, capturing and uploading geospatial data and timestamped photographs directly into the system database. At this juncture, the system presents a critical decision node: "Sample collection?". If the inspector determines that samples are not required based on the facility's risk profile, the workflow bypasses the laboratory phase, and the inspector executes the "Submit the Inspection Report" action directly to the central authority. Conversely, if samples are mandated, the inspector initiates the "Send for Lab test and Payment Request" protocol. The system calculates this secondary payment based precisely on the number of samples extracted.

Once the laboratory payment is cleared, the state shifts to the accredited testing facility, triggering the "Receive Samples" and "Test and Submit Lab Report" functions. The laboratory analyst uploads the PDF based test report. The central authority then reaches a validation node: "Lab test done?" and "Lab test Payment done?". The system is hardcoded to "Hold results until payment is completed," ensuring financial compliance before regulatory advancement.

Following full payment and data aggregation, the system initiates a highly structured, multi-tiered approval hierarchy. The complete digital dossier is sequentially forwarded to the Assistant Director, who executes the "Receive Reports" and "Give Recommendation" actions. The recommended file then escalates to the Director, and ultimately to the Director General for final statutory authorization.

At the "Results approved?" decision node, a negative determination routes the system to "Reject the application," generating an automated notification detailing the specific reasons for rejection to the applicant. A positive determination triggers the system to "Issue the Certificate of Registration," granting a statutorily defined two-year validity period while dynamically generating a unique registration number linked to the applicable calendar year.

Phase 02: Post-Registration Surveillance and Periodic Reviews

The issuance of a registration certificate does not conclude the system's involvement with the facility; rather, it initiates a continuous, automated surveillance loop. The system architecture incorporates a sub-module designed to mandate and track actions defined as "Conduct site inspections time to time and upload results and recommendations" by authorized officers in the relevant geographical area.

These periodic inspection reports are systematically ingested into the database, triggering a "Review of periodically site inspection results" by the central authority. The system presents a decision node: "Further actions required?".

If the periodic audit reveals compliance deviations, the system enables the central authority to "Take appropriate follow-up actions by assigning statuses." These dynamic system statuses include "Revisit Required," "Further Inspection," "Correction Period Granted," or in severe cases, "Certificate Cancellation."

The system tracks the remediation timelines associated with these statuses. If the subsequent "Outcome is successful?", the system executes the "approve the certification and continue its validity period" function. If the manufacturer fails to remediate the critical non-conformities, the system automatically shifts the facility's state to "certificate revocation," immediately nullifying the entity's legal capacity to operate and locking their ability to generate transport permits.

Phase 03: Registration Renewal Workflow

Statutory regulations dictate strict validity periods for manufacturing and storage certificates, necessitating a robust, automated renewal mechanism. To minimize administrative friction, the software allows the applicant to "Login to the Profile" and "Fill Renewal Application with required documents," featuring a critical system capability: the "ability to reuse previously submitted documents where applicable." If structural plans or factory layouts remain entirely unchanged, the applicant simply verifies their accuracy rather than executing redundant uploads.

Upon submission of the renewal request, the central authority executes the validation protocol, leading to the "Payment Request - Renewal fee + Inspection fee (differ based on the distance)". Following payment, the system routes the file to the district inspector to "Conduct site inspection and generate site inspection report (uploading site visit photos, if required)." The subsequent workflow perfectly mirrors the new registration process, navigating the "Sample collection?" decision node, the potential routing for laboratory testing and secondary sample payments, and the mandatory holding of results until financial clearance.

The multi-tiered approval hierarchy remains engaged for renewals, requiring progressive sign-offs from the Assistant Director, Director, and Director General. At the final "Results approved?" node, an affirmative decision prompts the system to "Issue the Updated Certificate," seamlessly extending the facility's legal operating window while preserving the immutable historical audit trail of previous years' compliance data.

Phase 04: Registration Amendments Workflow

Operational realities within the industrial sector dictate that manufacturers will frequently undergo structural, administrative, or processing modifications. The software architecture explicitly accommodates a dedicated Registration Amendment workflow to handle these inevitabilities. Regulations mandate that any material changes to the initially provided organizational information must be updated within the central system within a strict fourteen-day window.

If a registered saltern/factory undergoes any changes during the validity period of the certificate (e.g., ownership, area, operations, etc.), the license holder must inform the relevant authority and obtain an updated/amended certificate. Failure to notify such changes may result in cancellation of the license.

The applicant initiates this process by authenticating into their profile and triggering the "Requests Amendment" function. The system must capture specific categorical changes, explicitly identifying alterations to "location/address/owner change" or facility layout expansions. The applicant is required to "Upload Supporting Documents" pertinent specifically to the requested modification.

The central authority executes the "Receive the Amendment request" and "Review the documents" functions. Depending on the severity and nature of the amendment, the system's business logic allows the authority to determine the necessity of physical verification, triggering the "Conduct Site Inspections/ Sample testing (if required)" node. Following this review, an initial "Approved?" decision is made. A negative determination results in a direct "Reject" action. A positive determination requires the system to "Forward to DG for Final Authorization."

The Director General executes the "Receive the request for Amendment Approval" and "Review the documents and Provide approval" functions. The final "Approved?" decision node dictates the ultimate system state. Rejection generates a notification to the applicant, while approval triggers the system to "Issue Updated Certificate (if required) or change company profile," ensuring the master database perfectly reflects the current, legal reality of the manufacturing entity.

Phase 05: Multiple Brands Registration Workflow

The regulatory framework requires separate approvals for each registered salt brand/product category. To support operational scenarios where a single registered factory or saltern produces multiple commercial salt brands, the system shall provide a streamlined Multiple Brand Registration workflow based on a parent-child registration structure.

An existing, fully registered Factory/Saltern owner shall log into the portal and initiate a “New Brand Registration Request” by selecting the already approved and active Company/Factory Profile. The system shall avoid redundant collection of previously verified facility-level information, such as factory/saltern details, licenses, inspection records, and supporting registration documents.

The applicant shall then provide the new brand-specific information, including:

- Brand name
- Salt category (e.g., Iodized Food Grade Salt / Non-Iodized Salt)
- Product label/artwork
- Packaging details
- Any additional brand-specific documents

The system shall provide an option to reuse previously submitted and approved documents associated with the parent registration. If the applicant selects “Reuse Existing Documents,” the new brand application shall be linked to the already verified factory/saltern information, inspection reports, laboratory test records, and approvals maintained under the existing registration profile. If new or updated documents are required, the applicant shall be allowed to upload them separately.

Upon submission, the relevant authority shall review the secondary brand registration request, verify compliance with applicable regulatory requirements, and process the application accordingly.

Once approved, the system shall generate and issue a separate Registration Certificate/License for the new brand while maintaining its association with the primary Factory/Saltern registration profile. For example, a single registered factory may hold multiple valid brand registrations under the same parent facility

9.2 Use Case Modeling and Narratives

To effectively translate the complex, multi-actor operational workflows into precise, actionable software development requirements, the following use case models meticulously define the interactions between the human actors, external system interfaces, and the internal application boundaries.

UC01: Submit Application for Common Salt Manufacturing (Factory/Saltern)

Description: The comprehensive process by which a business entity or individual owner creates an authenticated profile and submits a highly detailed new registration application for either a salt manufacturing processing facility or a raw salt extraction Saltern.

Actors: Applicant (Manufacturer/Owner), System.

Pre-conditions: The Applicant has successfully navigated the public portal, executed the "Sign up to the system" function, verified their contact credentials (email), and possesses all digitized, legally certified documents required for submission.

Post-conditions: The comprehensive application payload is securely saved in the relational database, a unique, cryptographically secure application reference number is generated, and the workflow state is formally updated to "Pending Initial Validation".

Step	Action Flow
1	Applicant authenticates into the portal and selects the specific "New Registration" pathway, differentiating between a Factory or Saltern setup. System renders the appropriate dynamic digital application form, altering required fields based on the facility type selection.
2	Applicant inputs exhaustive Business Profile data, including Legal Name, Business Registration Number, Head Office Address, and detailed Director Board information. System executes real-time validation checks on field formats (e.g., ensuring alphanumeric integrity for BR numbers and standard length for contact details).
3	Applicant inputs specific Factory/Saltern Site Details, selecting the District, Divisional Secretariat, and Local Authority, while providing exact GPS coordinates.

4	Applicant specifies the precise product classification, explicitly selecting between "Iodized Food Grade Salt" or "Non-Iodized Salt," allowing for multiple selections if the facility handles both. System dynamically adjusts subsequent form fields, laboratory requirement flags, and the mandatory document upload checklist based on the statutory requirements for the selected categories.
5	Applicant uploads all mandated digital artifacts, including the Business Registration, Owner NIC, Local Authority Certificates, Site Plans, Factory Layouts, Process Flow Charts, and high-resolution Product Labels. System actively scans uploaded files for valid extensions (PDF, JPG, PNG), enforces maximum file size constraints, and verifies against basic malware signatures.
6	Applicant digitally executes the legal declaration confirming data accuracy and triggers the final "Submit" action. System commits the complete payload to the database. Generates the unique Application ID. Transitions the system state to "Pending Initial Validation" by the central authority.

UC02: Execute Preliminary Validation and Dynamic Payment Processing

Description: The process involving the central authority's initial review of the submitted digital dossier and the subsequent calculation and collection of the statutory registration and distance-based inspection fees via the government digital payment platform.

Actors: Central Authority Officer, Applicant, IPG, System.

Pre-conditions: Application is in the "Pending Initial Validation" state.

Post-conditions: The initial documentation is deemed legally sufficient, financial obligations are fully realized, a digital receipt is generated, and the workflow state transitions to "Pending District Inspection".

Step	Action Description
1	Central Officer accesses the administrative dashboard, opens the queued application, and executes the "Review and Validate the application" function, verifying document clarity and legal compliance. System enables the "Approve for Payment" or "Return for Correction" action buttons.
2	Central Officer Triggers the "Approve for Payment" action. The system algorithmically calculates the total required fee by aggregating the base registration cost with a variable inspection fee derived dynamically from the geographical distance between the central office and the facility's GPS coordinates. System dispatches a payment request notification to the Applicant.
3	Applicant accesses their user dashboard, reviews the fee breakdown, and clicks the "Proceed to Payment" portal link. Constructs a secure, encrypted financial payload (containing the Application ID, exact LKR amount, and return callbacks) and seamlessly redirects the user session to the IPG URL.
4	IPG presents an omnichannel suite of payment modalities, including CEFTS, SLIPS, credit/debit card interfaces, and mobile financial applications. The FCAU system enters a listening state, awaiting asynchronous transaction confirmation.
5	Applicant completes the stringent authentication protocols and finalizes the financial transfer within their chosen banking environment. The IPG processes the clearing and settlement protocols.
6	IPG transmits an asynchronous, cryptographically signed webhook confirming the successful realization of funds to the FCAU system. Intercepts the webhook payload, rigorously verifies the cryptographic signature against stored public keys to prevent spoofing, and updates the internal financial ledger.
7	System detects successful ledger update.

	Automatically generates a downloadable PDF receipt. Transitions the overarching application state to "Pending District Inspection." Dispatches routing notifications to the relevant district Food and Drugs Inspector.
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UC03: Execute Field Inspection and Laboratory Testing Protocol

Description: The critical sequential review of the physical facility by localized health authorities, the potential extraction of physical samples, and the subsequent analytical testing by accredited laboratory personnel to ensure strict adherence to safety and chemical standards.

Actors: District Food & Drugs Inspector, Laboratory Analyst, System.

Pre-conditions: Application state is "Pending District Inspection". The initial payment has been successfully cleared.

Post-conditions: Field data is aggregated, lab results are computationally verified against statutory limits, and the application is routed to the multi-tiered approval hierarchy.

Step	Action Description
1	District Inspector travels to the facility, accesses the mobile-responsive system interface, and conducts the physical audit. Inputs findings into the digitized "Food Premises Grading Form" and uploads real-time, geotagged photographs. Saves the inspection data payload. Presents the critical decision node: "Sample collection required?".
2	District Inspector determines that physical salt samples are necessary for chemical and microbiological validation and triggers the "Send samples for Lab test" function, indicating the exact number of samples extracted. System transitions the specific sample tracking sub-state to "Awaiting Lab Receipt". Generates a secondary, automated "Payment Request" directed to the Applicant, calculating the fee proportionally based on the sample count.
3	Applicant executes the secondary payment via the IPG integration for the laboratory testing fees. System verifies the payment utilizing the identical webhook logic detailed in UC02. Updates the laboratory tracking module to "Payment Cleared - Testing Authorized".

4	Lab Analyst physically receives the samples, logs into the laboratory portal, and acknowledges receipt. Executes the required assays (e.g., Moisture, NaCl, Iodine, Salmonella) and upload the final report to the system.
5	System detects the completion of the laboratory report and the successful clearing of all associated fees. Transitions the overarching application state to "Pending Assistant Director Review".

UC04: Multi-Tiered Authorization and Certificate Issuance

Description: The final, highly structured administrative workflow requiring sequential review and authorization by senior public health officials, culminating in the cryptographic generation of the legal registration certificate.

Actors: Assistant Director, Director, Director General, System.

Pre-conditions: The application state is "Pending Assistant Director Review". All field inspections and laboratory results (if applicable) are aggregated and accessible.

Post-conditions: The facility is legally registered, and the applicant receives a verifiable digital certificate possessing a two-year validity period.

Step	Action Description
1	Assistant Director accesses the queue, comprehensively reviews the aggregated digital dossier, inspection reports, and laboratory analytics. Executes the "Give Recommendation" function. System logs the officer's action and timestamp. Transitions the state to "Pending Director Review".
2	Director accesses the escalated queue, verifies the Assistant Director's recommendation, and executes the secondary "Provide Approval" function. System logs the officer's action. Transitions the state to "Pending Director General Authorization".
3	Director General accesses the final authorization queue, reviews the complete audit trail of recommendations, and makes the definitive statutory decision at the "Results approved?" node.

	If rejected: the system routes to the "Reject the application" state, requiring the DG to input mandatory rejection reasons, which are emailed to the applicant.
4	Director General selects the affirmative approval option to authorize the facility. The system algorithmically generates the formal Certificate of Registration (aligning with Schedule II, III, or V based on the salt category) or provide the facility to upload the certificate with a verifiable QR code. Calculates the expiration date precisely two years from the moment of issuance. Transitions the facility state to "Active Registration". Dispatches the certificate to the Applicant's portal and email.

UC-05: Process Multiple Brand Registrations

Description: Allows an existing registered salt factory/saltern owner to efficiently register an additional commercial salt brand using the already verified facility, inspection, and registration information maintained within the system.

Actors: Applicant / Factory Owner, Relevant Authority (FCAU), System

Pre-conditions: The applicant has a fully registered and active Factory/Saltern Profile within the system.

Post-conditions: A separate Registration Certificate/License is issued for the new brand while maintaining linkage to the parent Factory/Saltern Profile.

Step	System Action / User Interaction
1	Applicant logs into the system and selects "Request New Brand Registration" from the dashboard by choosing the existing registered Factory/Saltern Profile.
2	Applicant enters the specific details related to the new brand, such as Brand Name, Salt Category (e.g., Iodized Food Grade Salt / Non-Iodized Salt), packaging details, and uploads the relevant brand-specific documents and label/artwork.
3	System presents an option asking whether the applicant wishes to reuse previously submitted and approved documents from the parent registration profile.

4	Applicant selects “Yes” to reuse existing factory/salern registration details, inspection reports, laboratory test records, and previously approved supporting documents.
5	System links the existing approved records from the parent profile to the new brand registration request and prompts the applicant to submit the application.
6	FCAU receives the request and reviews the newly submitted brand-specific details, labels, and packaging materials to ensure compliance with applicable food labeling and regulatory requirements.
7	Upon approval, the system generates and issues a separate Registration Certificate/License for the new brand while maintaining its relationship with the original Factory/Salern registration profile.

9.3 Data Dictionary and Entity-Relationship Architecture

A highly normalized, robust relational database schema is absolutely critical to ensuring the software module can support complex, high-velocity queries, immutable audit tracking, and seamless regulatory reporting. The following data dictionaries outline the primary entities and field specifications derived directly from the FCAU application requirements, statutory guidelines, and flow diagram logic.

Entity: Business_Entity_Profile

This foundational entity captures the core demographic, legal, and tax-related details of the manufacturing organization.

Field Name	Data Type	Constraint	Description / Validation Rule
Entity_ID	UUID	Primary Key	System-generated, globally unique identifier.
Legal_Company_Name	Varchar(150)	Mandatory	Official registered name of the manufacturing entity.

Business_Registration_No	Varchar(50)	Mandatory	Official BR Number. Must enforce database-level uniqueness.
Date_of_Incorporation	Date	Mandatory	Historical date the company was legally established.
TIN_Certificate_No	Varchar(20)	Mandatory	Taxpayer Identification Number issued by Inland Revenue.
VAT_Registration_No	Varchar(20)	Optional	Value Added Tax certificate number, if applicable.
Registered_Head_Office	Text	Mandatory	Complete physical address of the corporate headquarters.
Primary_Contact_Phone	Varchar(15)	Mandatory	Strict Regex validation for standard Sri Lankan telecommunication formats.
Authorized_Owner_Name	Varchar(100)	Mandatory	Full legal name of the primary stakeholder or Managing Director.
Authorized_Owner_NIC	Varchar(12)	Mandatory	Regex validation ensuring compliance with both Old (9 digits + V/X) and New (12 digits) National Identity Card formats.

Entity: Operational_Facility_Details

This entity models the physical, geographical site under regulation, differentiating between raw extraction environments and refined processing centers.

Field Name	Data Type	Constraint	Description / Validation Rule
Facility_ID	UUID	Primary Key	System-generated, globally unique identifier.
Entity_ID	UUID	Foreign Key	Relational link mapping the facility back to the Business_Entity_Profile.

Facility_Classification	Enum	Mandatory	Allowed Values:.
Approved_Salt_Category	Enum	Mandatory	Allowed Values:.
Administrative_District	Enum	Mandatory	Dropdown selection locked to the 25 recognized administrative districts of Sri Lanka.
Divisional_Secretariat	Enum	Mandatory	Cascading dropdown dynamically filtered based on the selected Administrative_District.
Geospatial_Latitude	Decimal(9,6)	Mandatory	Exact geographical coordinate enabling spatial mapping and distance-based fee calculation.
Geospatial_Longitude	Decimal(9,6)	Mandatory	Exact geographical coordinate enabling spatial mapping and distance-based fee calculation.
Local_Authority_Name	Varchar(100)	Mandatory	Explicit name of the governing Pradeshiya Sabha, Urban Council, or Municipal Council.

Entity: Regulatory_Document_Repository

The software system must implement large-scale, highly secure object storage capable of managing the mandatory compliance artifacts. All documents referenced herein must be encrypted at rest to preserve commercial confidentiality.

Field Name	Data Type	Constraint	Description / Validation Rule
Document_ID	UUID	Primary Key	Unique document reference tracking hash.
Facility_ID	UUID	Foreign Key	Relational link binding the document to a specific operational facility.

Document_Category	Enum	Mandatory	Allowed Values:.
Encrypted_File_URI	Varchar(255)	Mandatory	Secure, authenticated routing link to the AWS S3 (or equivalent infrastructure) storage bucket.
System_Ingestion_Time	Timestamp	Auto	Immutable timestamp recording the exact moment of upload.
Authority_Verification_Status	Enum	Default 'Pending'	Allowed Values:.. Actively updated by central FCAU or SLSI officers during UC02.

Entity: Analytical_Laboratory_Results

This entity is undeniably the most critical component for ensuring public health safety, as it computationally enforces the stringent parameters of the Food (Iodization of Salt) Regulations.

Field Name	Data Type	Constraint	Description / Validation Rule
Analytical_Test_ID	UUID	Primary Key	System-generated identifier for the specific laboratory assay suite.
Facility_ID	UUID	Foreign Key	Relational link to the facility from which the samples were physically extracted.
Sample_Extraction_Date	Date	Mandatory	Precise date the District Inspector collected the physical sample from the site.

Moisture_Percentage	Decimal(4,2)	Mandatory	Automated System Flag: Value must be strictly < 6.00 % by total weight.
Insoluble_Matter_Percentage	Decimal(4,2)	Mandatory	Automated System Flag: Value must be strictly < 0.50 % calculated on a dry basis.
NaCl_Purity_Content	Decimal(5,2)	Mandatory	Automated System Flag: Value must be strictly >= 97.00 % calculated on a dry basis.
Soluble_Matter_Percentage	Decimal(4,2)	Mandatory	Automated System Flag: Value must be strictly < 2.00 % (specifically excluding the NaCl content).
Iodine_Concentration	Decimal(5,2)	Conditional	Automated System Flag: Active for 'Iodized Food Grade' category ONLY. Value must fall strictly between 15.00 and 30.00 mg/kg.
Microbiology_Salmonella	Enum	Mandatory	Allowed Values:. Algorithmic requirement: Must be 'Not_Detected'.
Microbiology_E_Coli	Enum	Mandatory	Allowed Values:. Algorithmic requirement: Must be 'Not_Detected'.
Computed_Overall_Status	Enum	Auto-Calculated	Allowed Values:. The system's logic engine calculates this definitively based on whether all individual threshold

			parameters have been successfully met.
Lab_Report_URL	URL	Conditional	The secure system path to the formalized PDF certificate uploaded by authorizing officer.

10. Bottled Water Manufacturing and Registration Management Module

The Bottled Water Manufacturing and Registration Management Module represents a highly sophisticated, multi-tiered digital regulatory subsystem designed to govern the entirety of the commercial bottled water lifecycle within Sri Lanka. Operating under the direct jurisdiction of the FCAU, this system is engineered to replace legacy manual processes with a centralized, state-machine-driven software architecture. The primary mandate of this digital transformation is to ensure that all commercially available bottled and packaged water conforms to the stringent public health directives established by the Food Act and the subsequent Food (Bottled or Packaged Water) Regulations.

10.1 Comprehensive Workflow Elaboration

The core of the Bottled Water Manufacturing and Registration Management Module is an advanced workflow orchestration engine that models the precise procedural steps, actor responsibilities, and decision gateways outlined in the FCAU's official operational flowcharts. The architecture requires tracking applications across a distributed network of human actors, including the Manufacturer, FCAU administrative staff, Authorized Field Officers (PHI, MOH, FDI), external Laboratory personnel, SLSI auditors, the Deputy Director General (DDG), and the ultimate approving authority, the Director General (DG).

Phase 01: Initial Bottled Water Registration Workflow

The initial registration lifecycle represents the most rigorous and complex state machine within the module, requiring parallel processing and staged compliance verifications. The process commences with the Manufacturer accessing the public-facing portal to sign up, authenticate, and create a comprehensive company profile. Following profile creation, the user navigates to

the registration interface to complete the digital application and upload all mandatory statutory documents, including Google Map locations, survey plans marking the extraction source, approved building layouts, process flow charts, and organizational hierarchies. A client-side validation gateway actively prevents submission until all mandatory files are successfully attached to the digital dossier.

Upon successful document attachment, the system calculates the requisite fees and triggers a payment request. Crucially, the financial logic engine must calculate a composite total comprising the fixed registration fee (Rs. 50,000) alongside a variable inspection fee that fluctuates based on the geographic distance to the manufacturing site.

Following the secure realization of this payment, the FCAU receives the application for preliminary administrative review. Once validated, the system bifurcates the workflow, simultaneously dispatching the dossier to the Sri Lanka Standards Institution (SLSI) and generating automated notifications for Authorized Officers to conduct factory visits. The designated area Public Health Inspector (PHI), Medical Officer of Health (MOH), and Food and Drugs Inspector (FDI), alongside an FCAU inspection team, conduct rigorous site visits, capturing photographic evidence and compiling digital inspection reports directly into the system. Concurrently, Authorized Officers collect water samples and forward them to approved external or state laboratories for extensive chemical and microbiological testing.

The external Labs receive the physical samples, conduct the required analyses, and submit the digitized lab reports back into the central FCAU platform. Simultaneously, the SLSI reviews the application copies and conducts a primary site visit. Based on this initial audit, the SLSI uploads a Provisional Recommendation valid for one year. The central system then aggregates the field inspection reports, the SLSI provisional recommendation, and the laboratory results, forwarding the complete package to the Assistant Director or Deputy Director General (DDG) for executive review.

The DDG provides a formal recommendation and escalates the dossier to the Director General (DG) for final authorization. The system enforces a mandatory secondary payment gateway check at this juncture, holding final results until the applicant settles any outstanding laboratory testing fees. If the DG rejects the application based on the aggregated data, the system triggers an automated notification detailing the reasons for rejection. If approved, the system generates and issues a Certificate of Registration valid for one year, explicitly linked to the SLSI's provisional recommendation.

The system architecture continues to monitor the facility post-issuance. Within the first year, the SLSI conducts a second mandatory site visit. Upon the system receiving the SLSI's Full Recommendation, the DG issues the final registration certificate, extending the validity for the

subsequent two years, culminating in a total three-year lifecycle. The workflow engine also schedules periodic, unannounced site inspections. The results of these periodic reviews are fed back into a decision gateway. If anomalies are detected, the system triggers follow-up actions, assigning statuses such as "Revisit Required," "Correction Period Granted," or initiating the "Certificate Cancellation" protocol. If the manufacturer successfully remediates the issues, the certification continues its validity period; otherwise, the system executes a formal certificate revocation, updating the public registry accordingly.

Phase 02: Registration Renewal Workflow

To guarantee persistent regulatory compliance, the module enforces a strict renewal workflow as the three-year registration lifecycle approaches expiration. The system initiates this process by enabling the Manufacturer to access the renewal interface via their existing profile. To optimize the user experience and reduce database redundancy, the software architecture features a document reference engine that allows applicants to actively reuse previously submitted, non-expiring technical documents (such as fixed factory layouts or static survey plans) while requiring the mandatory upload of time-sensitive materials, such as renewed local authority licenses.

Upon submission of the renewal application, the system calculates and requests the composite payment, which again combines the base renewal fee with the dynamically calculated, distance-based inspection fee. Following payment realization, the FCAU reviews the application and dispatches notifications to the Authorized Officers for fresh factory inspections and water sampling.

The samples are routed to the approved Labs, which independently test the raw and finished water and submit the analytical reports back to the central platform. The system aggregates the updated inspection reports and lab data, forwarding them to the DDG for recommendation and the DG for final approval. The workflow mirrors the initial registration's strict gateway logic, holding final approval until all lab testing fees are verified as paid. Upon the DG's approval, the system updates the digital certificate's expiration date, initiating the next multi-year validity period while maintaining the automated schedule for ongoing periodic site inspections and potential revocation pathways if compliance fails during the new term.

Phase 03: Amendments Management Workflow

The module encompasses a dedicated subsystem to handle administrative and operational modifications to an actively registered facility, such as changes to the physical location, the

registered address, or the legal ownership structure. The Manufacturer logs into their profile and selects the specific amendment request type, subsequently uploading the necessary supporting legal or structural documentation.

The FCAU receives the amendment request and conducts a preliminary review of the attached files. Depending on the nature of the amendment, the system presents a decision gateway allowing the FCAU to determine if a physical site inspection or new sample testing is required. For instance, a minor administrative address change may bypass field verification, whereas a change in the factory's physical location or processing infrastructure mandates a full physical audit.

Following the execution of any required inspections, the workflow routes to an initial approval node. If the FCAU rejects the modification, the workflow terminates, and the manufacturer is notified. If approved at the preliminary stage, the system forwards the dossier to the DDG for executive review and subsequently to the DG for final authorization. Upon the DG's final approval, the system executes the necessary database updates, automatically reflecting the changes in the overarching company profile and issuing an updated, revised Certificate of Registration if the modification alters the data printed on the public-facing certificate.

Phase 04: Multiple Brands Registration Workflow

The regulatory framework mandates that approvals must be obtained separately in respect of each specific source, premises, and brand. To accommodate market realities where a single compliant facility produces multiple distinct commercial labels, the system implements a highly efficient Multiple Brands workflow based on a parent-child data architecture.

An existing, fully registered Manufacturer logs into the portal and initiates a "New Brand Registration Request," explicitly selecting their already verified and active "Existing Company Profile." The system bypasses the redundant collection of facility-level data (such as factory layouts, WRB approvals, and land deeds) and directs the user to enter the new brand-specific details, primarily focusing on the upload of the new commercial labels and packaging materials.

A critical decision gateway is presented to the user: "Reuse Existing Documents?" If the user selects "Yes," the system's relational database links the new brand application directly to the previously verified water source, lab tests, and SLSI audits associated with the parent company profile. If the user selects "No," the system prompts the upload of new, specific technical documents. Upon submission of this streamlined application, the FCAU receives the secondary brand request, verifies the label compliance against the Food (Labelling and Advertising) Regulations, and processes the Bottled Water Registration. The system concludes the workflow by generating and issuing a distinct, new Certificate of Registration specifically for the secondary

brand, while maintaining its hierarchical linkage to the single, overarching Company Profile (e.g., Facility X holds Certificate A for Brand A, and Certificate B for Brand B).

10.2 Use Case Modeling and Narratives

The subsequent use case models formalize the interactions between the defined human actors, the external integration points, and the core software system, ensuring that all business rules and procedural handoffs are explicitly captured for development.

UC-01: Execute Initial Registration and Provisional Issuance

Description: The end-to-end orchestration of capturing facility data, processing variable payments, aggregating field and lab reports, coordinating SLSI provisional audits, and securing final executive approval for a one-year temporary registration.

Actors: Manufacturer, FCAU, Authorized Officers, Labs, SLSI, DDG, DG, System

Pre-conditions: Manufacturer has authenticated, possesses necessary legal deeds, WRB approvals, and the proposed water source meets SLS standards.

Post-conditions: A one-year provisional Certificate of Registration is issued, and the facility is placed on the periodic inspection schedule.

Step	Action Description
1	Manufacturer authenticates into the system, navigates to the application interface, and populates mandatory fields regarding the firm, source location, and product type (Mineral vs. Drinking Water).
2	System enforces validation rules, including checking that the numeric entry for land_extent is equal to or greater than 160 perches.
3	Manufacturer uploads all mandatory attachments: GIS coordinates, survey plan, approved factory layout, flow chart, label specimens, and medical reports.
4	System verifies attachment completeness. Calculates the base registration fee (Rs. 50,000) plus the variable, distance-based inspection fee and generates a payment request gateway link.
5	Manufacturer executes the transaction via the integrated payment gateway.

6	FCAU receives the submitted dossier, performs an initial administrative review, and triggers the parallel evaluation pathways.
7	System simultaneously routes necessary documentation to the SLSI portal and generates inspection task notifications for the respective PHI, MOH, FDI, and FCAU inspection teams.
8	Authorized Officers conduct physical site inspections, capture geolocated photographic evidence, and submit structured inspection reports into the system.
9	Labs receive physical water samples, conduct chemical and microbiological analyses, and input the results or upload PDF report directly into the system's Laboratory Coordination module.
10	SLSI conducts the first physical site visit, evaluates compliance against SLS, and uploads a Provisional Recommendation (valid for one year) to the system.
11	System aggregates all submitted inspection reports, lab results, and the SLSI recommendation into a unified executive dossier and forwards it to the DDG.
12	DDG reviews the aggregated compliance data and provides a formal recommendation, escalating the record to the DG.
13	System halts final approval until a secondary financial verification confirms that the Manufacturer has successfully paid all external Laboratory testing fees.
14	DG upon verification of lab payments and satisfactory reports, grants final executive approval for the provisional license.
15	System generates the official Certificate of Registration with a 1-year expiration date, issues the document to the Manufacturer's portal, and activates the chron-job for the subsequent 2nd SLSI site visit.

UC-02: Manage Registration Renewal and Periodic Compliance

Description: Facilitates the triennial renewal of an active registration, allowing the reuse of static documents while enforcing new lab testing and field inspections to verify ongoing compliance

Actors: Manufacturer, FCAU, Authorized Officers, Labs, DDG, DG, System

Pre-conditions: The Manufacturer holds an active or soon-to-expire full Certificate of Registration.

Post-conditions: The registration validity is extended for a new term, or the certificate is formally revoked due to compliance failures.

Step	Action Description
1	System detects impending expiration of a final registration and dispatches automated alerts to the Manufacturer.
2	Manufacturer initiates the renewal workflow. Selects options to link and reuse previously submitted static documents (e.g., building layout) and uploads newly required time-sensitive files (e.g., annual licenses).
3	System calculates the composite renewal and distance-based inspection fee, prompting payment.
4	FCAU reviews the renewal submission and dispatches notifications for fresh field inspections.
5	Authorized Officers / Labs execute physical inspections and fresh water sampling. Labs process samples and upload updated chemical and microbiological compliance data with report.
6	System evaluates new lab data against statutory limits. Aggregates field reports and forwards the dossier to the DDG and subsequently the DG.
7	DG reviews the continuous compliance history and the new data, granting approval for the renewal.
8	System updates the certificate validity period. Continues to execute the background periodic inspection loop.
9	(Exception Flow) If a periodic inspection or renewal lab test fails, system flags the facility. FCAU assigns a remediation status (e.g., Correction Period). If remediation fails, system executes Certificate Revocation protocol.

UC-03: Process Multiple (maximum two) Brand Registrations

Description: Allows an established manufacturer to efficiently register an additional commercial brand utilizing the existing, verified infrastructure and water source data

Actors: Manufacturer, FCAU, System

Pre-conditions: The Manufacturer has a fully registered and active parent Company Profile within the system.

Post-conditions: A unique Certificate of Registration is issued for the new brand, hierarchically linked to the parent facility.

Step	System Action / User Interaction
1	Manufacturer authenticates and selects "Request New Brand Registration" from the dashboard, explicitly selecting their existing Company Profile.
2	Manufacturer inputs the specific details for the new brand (e.g., Brand Name, product range) and uploads the brand-specific documents (e.g., agreement, intellectual property rights, etc) , crucially including two specimens of the new commercial label.
3	System presents a decision gateway asking if the user wishes to reuse existing technical documents from the parent profile.
4	Manufacturer selects "Yes" to link existing facility and source documents, bypassing redundant data entry.
5	System links the document UUIDs from the parent profile to the new brand application and prompts for the application submission.
6	FCAU receives the streamlined request. Evaluates the newly uploaded label specimens for strict compliance with the Food (Labelling and Advertising) Regulations, ensuring no medical claims or misleading graphics are present.
7	FCAU approval, generates a distinct Certificate of Registration for the new brand (e.g., Certificate B) while maintaining relational database links to the original Company Profile.

10.3 Data Dictionary and Entity-Relationship Architecture

A highly structured, normalized relational database schema is paramount to enforcing the rigorous regulatory logic outlined in Schedule I (Application particulars) and Schedule III (Physical, Chemical, and Microbiological maximum permissible limits) of the regulations. The data architecture relies on a parent-child hierarchy to manage the complex relationships between physical facilities, geological water sources, and commercial brands.

Entity: Facility_Registration (The Parent Node)

This entity stores the immutable and structural details of the manufacturing premises and the legal entity operating it. It serves as the parent record to which multiple brands and water sources can be linked.

Field Name	Data Type	Constraint	Description
facility_uuid	UUID	PRIMARY KEY	Unique system identifier for the physical plant.
unit_name	VARCHAR(255)	NOT NULL	The formal name of the manufacturing unit.
address_street	VARCHAR(255)	NOT NULL	Complete physical address including street boundaries.
geo_coordinates	POINT	NOT NULL	Stored geospatial data extracted from the required Google Map upload, used for calculating the distance-based inspection fee.
firm_type	ENUM	NOT NULL	Restricted to 'Public', 'Private_Ltd', or 'Proprietary'.
managing_director	VARCHAR(255)	NOT NULL	Full name of the primary executive officer.
auth_person_name	VARCHAR(255)	NOT NULL	Name of the individual authorized to legally bind the firm in the application.

land_extent_perches	DECIMAL(8,2)	$\geq$	Strict statutory enforcement: The software must reject any submission where the industrial land is less than 160 perches.
ownership_tenure	ENUM	NOT NULL	'Owned' or 'Leased'. Determines which type of legal deed or 5-year lease agreement must be uploaded to the Document_Repository.
system_status	ENUM	NOT NULL	The current state of the facility (e.g., DRAFT, PENDING_INSPECTION, PROVISIONAL_ACTIVE, FULL_ACTIVE, REVOKED).

Entity: Water_Source_Details

This entity isolates the geological, geographical, and regulatory data concerning the specific point of water extraction, allowing multiple brands to reference a single validated source.

Field Name	Data Type	Constraint	Description
source_uuid	UUID	PRIMARY KEY	Unique identifier for the extraction point.
facility_uuid	UUID	FOREIGN KEY	Relational link to the parent manufacturing facility.
source_type	ENUM	NOT NULL	Strictly limited by regulation to 'Spring', 'Tube_Well', or 'Dug_Well'. Surface water is programmatically excluded.

distance_to_plant	DECIMAL(8,2)	NOT NULL	Numeric distance from the extraction point to the bottling facility, required by Form A.
wrb_approval_ref	VARCHAR(100)	NULLABLE	The alphanumeric reference code verifying compliance with the Water Resource Board

Entity: Brand_Certification (The Child Node)

This table manages the one-to-many relationship allowing a single verified Facility_Registration to legally produce and market multiple distinct commercial products.

Field Name	Data Type	Constraint	Regulatory / Business Logic Rationale
brand_uuid	UUID	PRIMARY KEY	Unique identifier for the specific commercial brand.
facility_uuid	UUID	FOREIGN KEY	Links the brand back to the verified parent facility and its associated technical documents.
brand_name	VARCHAR(255)	NOT NULL	The commercial name printed on the label.
water_category	ENUM	NOT NULL	'Natural_Mineral_Water' or 'Drinking_Water'. This selection dictates which set of maximum permissible limits the rules engine applies during lab test evaluation.
cfa_registration_no	VARCHAR(100)	UNIQUE	The official, legally mandated registration number issued by the Chief Food Authority that must be prominently displayed on the retail label.

sls_standard_no	ENUM	NULLABLE	Tracks adherence to 'SLS_894' for drinking water or 'SLS_1038' for natural mineral water.
issue_date	DATE	NOT NULL	The timestamp of certificate generation.
expiry_date	DATE	NOT NULL	The calculated expiration date (1 year for provisional, 3 years for full) driving the automated renewal notification system.

Entity: Laboratory_Compliance_Log

This critical entity captures structured data from the external laboratories, serving as the data source for the automated rules engine that determines pass/fail compliance against the Schedule III limits.

Field Name	Data Type	Constraint	Regulatory / Business Logic Rationale
test_uuid	UUID	PRIMARY KEY	Unique identifier for the testing event.
facility_uuid	UUID	FOREIGN KEY	Link to the facility being tested.
sample_state	ENUM	NOT NULL	Identifies if the sample is 'RAW_SOURCE_WATER' or 'FINISHED_BOTTLED_PRODUCT'.
testing_authority	ENUM	NOT NULL	Must be an approved entity: 'ITI', 'GOVT_ANALYST', 'MRI', or 'CITY_MICRO_LAB'.
turbidity_ntu	DECIMAL(5,2)	<=	Automated flag if value exceeds 5.0 NTU for either water category.
tds_mgl	DECIMAL(8,2)	NOT NULL	Automated flag if Total Dissolved Solids exceeds 1500.0 mg/l (Mineral) or 1000.0 mg/l (Drinking).

ph_level	DECIMAL(4,2)	6.5 <= x <=	Automated flag if Drinking Water falls outside the 6.5 to 8.5 range.
arsenic_mgl	DECIMAL(6,4)	<=	Critical health parameter; automated flag if Arsenic exceeds 0.01 mg/l.
lead_mgl	DECIMAL(6,4)	<=	Critical health parameter; automated flag if Lead exceeds 0.01 mg/l.
ecoli_status	BOOLEAN	EXPECT FALSE	Binary compliance check. Any result of TRUE (Present) triggers an immediate critical rejection.
coliforms_status	BOOLEAN	EXPECT FALSE	Binary compliance check. Must be FALSE (Absent).

11. Risk based Food Safety Inspection via Risk Analysis Framework

The modernization of national food control infrastructures represents a critical evolution from reactive, compliance-oriented enforcement toward a proactive, predictive, and science-based risk management paradigm. To operationalize this paradigm shift, the establishment of a specialized Risk Analysis Framework Module within the central Food Safety Inspection Management System is required. Functioning as the apex architectural component, this module represents the final integration layer, systematically unifying disparate data streams from public health inspectors, analytical laboratories, customs authorities, and educational platforms. This Software Requirements Specification outlines the exhaustive functional, technical, and architectural parameters required to engineer, integrate, and deploy this module. By codifying the internationally recognized principles of risk analysis into automated algorithmic workflows, the software empowers the Food Control Administration Unit (FCAU), Medical Officers of Health (MOH), and Public Health Inspectors (PHI) to dynamically allocate limited regulatory resources, prioritize high-risk establishments, and execute real-time, data-driven interventions that safeguard public health.

11.1 Theoretical Framework: The Tripartite Risk Analysis Engine

To ensure algorithmic integrity, the software architecture must faithfully replicate the scientific and regulatory principles of risk-based food inspection as delineated by the Codex Alimentarius Commission and the FAO. The system fundamentally distinguishes between a "hazard" defined as a biological, chemical, or physical agent with the potential to cause adverse health effects and a "risk," which is the mathematical probability and severity of that harm occurring within a specific population. The core risk engine evaluates these variables by digitizing the three interactive processes of the holistic risk analysis framework: **Risk Assessment, Risk Management, and Risk Communication.**

11.1.1 Automated Risk Assessment Protocols

The Risk Assessment component operates as the primary analytical processor within the software, executing a continuous, scientific evaluation of potential adverse health effects.

- The first step, Hazard Identification, involves the system continuously ingesting and aggregating data from national food surveys, import inspection logs, and field sampling results to pinpoint specific biological (e.g., *Salmonella*, *Listeria*), chemical (e.g., pesticide

residues, heavy metals), or physical agents present in the supply chain. Following identification, the system executes Hazard Characterization by querying to evaluate the severity of the potential adverse health effects, distinguishing between minor, self-limiting illnesses and severe, life-threatening conditions.

- Finally, Risk Characterization synthesizes these inputs, utilizing probabilistic models to estimate the overall likelihood and severity of the adverse effect.

11.1.2 Algorithmic Risk Management and Mitigation

Operating directly upon the outputs of the risk assessment engine, the Risk Management module translates scientific findings into actionable, prioritized regulatory policies. This module evaluates multiple policy alternatives and executes automated decision trees to select the most proportionate control measures based on the calculated risk severity. Within the software, this manifests as the dynamic adjustment of inspection frequencies, the automated triggering of product seizure protocols, and the generation of targeted compliance mandates. Furthermore, the management engine tracks the implementation and efficacy of these interventions over time, continuously feeding operational outcomes back into the assessment engine to refine future predictive models.

11.1.3 Integrated Risk Communication and Alert Telemetry

The Risk Communication module ensures the transparent, rapid, and interactive exchange of critical risk information among the system stakeholders. When the system detects a critical hazard threshold breach, automated notifications will trigger for relevant stakeholders.

11.1.4 Algorithmic Risk Categorization and Dynamic Inspection Scheduling

The operational core of the Risk Analysis Framework Module is its ability to objectively categorize Food Business Owners (FBOs) and dictate regulatory interaction schedules based on hard mathematical risk profiles rather than arbitrary calendar dates. The software utilizes a multifaceted algorithm that calculates a baseline risk score based on the inherent nature of the commodity handled, modified dynamically by the specific operational context and historical compliance record of the establishment.

Baseline Entity Risk Initialization

Upon the digital registration of a new food premises, the software immediately queries its categorization matrix to establish a baseline risk level. This matrix is strictly aligned with the regulatory guidelines established by the Food Control Administration Unit and the FAO, ensuring uniform application across all jurisdictions.

Risk-Ba sed No.	Premises Categorization (Risk-Based)	Inherent Hazard Profile	Baseline Risk Level
A1 / A2	Hotel, Resort, Restaurant, Cooked Food Establishment	Microbial contamination, temperature abuse, cross-contamination between raw and cooked items.	High
D / D1	Milk and milk product manufacturers, Ice cream, desserts	Contamination by pathogens, susceptibility to Aflatoxin in raw milk, non-food grade packaging.	High
E2	Street food vendors	Poor personal hygiene, unsafe water sources, inadequate waste disposal, ambient temperature exposure.	High
F	Bakery	Low moisture content products limiting microbial proliferation, high-temperature processing.	Low
G1	School canteens	Institutional vulnerability, prolonged storage of cooked food, targeted at susceptible child populations.	Medium
H2	Fish and Meat product stalls	High microbial load, rapid spoilage potential, cold chain failure, veterinary drug residues, heavy metals (mercury).	High
I1 / I2	Coconut oil mills, Edible oil manufacturers	Fungal infestation of copra leading to severe Aflatoxin formation, improper storage.	High

I4	Warehouses and Storage facilities	Moisture/temperature fluctuations, pest infestation, chemical contamination.	Medium
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This rigid classification ensures that an establishment dealing in highly perishable, pathogen-prone commodities like raw poultry or dairy is instantly flagged by the system for maximum regulatory oversight, regardless of its geographic location or operational scale.

11.1.5 Certification Driven Frequency Modulation

To incentivize industry led quality assurance and reduce the burden on the state inspectorate, the dynamic scheduling engine incorporates a frequency modulation algorithm tied directly to verifiable system certifications. The baseline inspection frequency for any "High Risk" establishment lacking certified food safety management systems is strictly programmed at a monthly interval. However, the software provides a secure portal for POs to upload cryptographically verifiable documentation proving adherence to recognized frameworks such as Hazard Analysis and Critical Control Points (HACCP), Good Manufacturing Practices (GMP), Good Agricultural Practices (GAP), or ISO 22000.

Upon the digital validation of these credentials by the central authority, the scheduling algorithm automatically recalibrates the compliance burden, reducing the mandatory inspection frequency for the high-risk establishment from every month to once every three months. Similarly, "Medium Risk" facilities holding valid certifications see their inspection intervals extended from every three months to every six months. The system actively tracks the expiration dates of these certificates; should a certification lapse without a verified renewal upload, the algorithm immediately revokes the extended interval and reverts the establishment back to the high frequency inspection schedule, automatically dispatching an alert to the assigned Public Health Inspector.

11.1.6 The Digital H-800 Compliance Assessment Architecture

The execution of the physical inspection is standardized through the digitization of the traditional Sri Lankan "Food Premises Grading Form," universally known as the H-800. The software transforms this paper based checklist into an interactive, context aware digital workflow accessible via the inspector's mobile responsive interface. The digital H-800 forces a rigorous, sequential evaluation across five primary operational domains: general environment, building infrastructure (including drainage and pest proofing), food handler hygiene (including

verification of medical fitness certificates), processing area controls (such as temperature logs and cross-contamination prevention), and food storage methodologies.

The application utilizes algorithmic determinism to eliminate subjective grading biases. As the inspector inputs observations, the system tallies the scores across the five domains, automatically calculating a cumulative percentage and assigning a definitive regulatory grade. Establishments scoring between 75% and 100% are awarded a Grade A, indicating good compliance; scores between 50% and 74% receive a Grade B; scores between 25% and 49% receive a Grade C, which triggers an automated reduction in the inspection interval to a monthly requirement until remediation is proven.

If the algorithmic calculation results in a score between 0% and 24% the establishment will receive a Grade D, this information is stored and maintained as history in the system. Only for further analysis purposes the information is being used, there is no impact to the Establishment or Food Business Owners (FBOs).

11.2 Use Case Modeling and Narratives

To precisely delineate the interaction between human actors, external systems, and the Risk Analysis Framework Module, the following highly detailed use case narratives document the sequential execution of critical regulatory procedures.

UC-01: Algorithmic Onboarding and Risk Initialization

Parameter	Operational Detail
System Actor	Medical Officer of Health (MOH) / Risk Calculation Engine
Triggering Event	A proprietor initiates a new premises registration inputting trade details, geocoordinates, and ownership data.
Execution Sequence	1. The MOH selects the operational category code corresponding to the food items inside the premises. The system processes this input against the central FAO categorization matrix.

	2. The Risk Calculation Engine evaluates the inherent hazard profile of the category, identifying vulnerabilities. Based on these factors, the engine deterministically assigns a baseline "High" risk level. 3. The software autonomously queries the document repository for verified quality management certifications (e.g., HACCP, GMP). Discovering no valid uploads, the scheduling algorithm applies the default maximum enforcement parameter. 4. The Dynamic Scheduling Module generates a recurring monthly inspection requirement for the establishment. 5. The system maps the establishment's GPS coordinates to the designated health division and automatically populates the assigned Public Health Inspector's digital itinerary with the initial baseline assessment task, flagged with a 14-day completion SLA.
Alternate Resolution	Should the proprietor upload a digitally signed, externally verifiable HACCP certification during the onboarding phase, Upon verification, the scheduling algorithm automatically modulates the inspection frequency, extending the interval from monthly to every three months, significantly reducing the regulatory burden while maintaining algorithmic risk fidelity.

UC-02: Automated Laboratory Hazard Detection and Escalation

Parameter	Operational Detail
System Actor	Notification Broker
Triggering Event	A formal chemical analysis is completed on a sample previously collected by a Food and Drug Inspector during a routine surveillance sweep.
Execution Sequence	1. The analyst submits the report to the system 2. The system parses the payload, matches the reference number to the digital Sample Register, and updates the record status to "Critical Non-Compliance". 3. The Risk Calculation Engine immediately overrides the manufacturer's current standing, locking their profile at an "Extreme Risk" threshold. 4. The Notification Broker assumes control, simultaneously triggering high-priority email alerts to the Chief Food Authority, the regional MOH, and the initiating inspector, detailing the batch numbers requiring immediate market seizure. 5. An automated legal dossier is compiled, aggregating the establishment's history, the digital chain of custody, and the certified lab result, preparing the framework for prosecution under the Food Act.

UC-03: Cross-Agency Import Rejection and Domestic Market Quarantine

Parameter	Operational Detail
System Actor	Customs Officer / FCAU Border Inspector
Triggering Event	A commercial shipping container arrives at the Port of Colombo.
Execution Sequence	1. The Sri Lanka Customs Single Window system processes the import declaration and queries the FCAU System's API for supplier risk profiles. 2. The risk engine identifies the foreign supplier as having a history of elevated mercury levels in previous shipments. The API returns a "Mandatory Hold and Test" directive back to the customs system. 3. An FCAU border inspector is dispatched, draws representative samples from the container, and logs the collection directly into the module, placing a digital quarantine lock on the consignment. 4. Subsequent laboratory integration confirms. The system automatically updates the consignment status to "Rejected at Border". 5. The system generates an automated notification to alert authority regarding the toxic shipment.

11.3 Data Dictionary and Entity-Relationship Architecture

The structural integrity and analytical power of the software depend entirely upon a meticulously engineered relational database schema. This schema must enforce strict referential integrity, standardize vast arrays of disparate data points, and maintain immutable historical ledgers to support predictive modeling and withstand legal scrutiny. The following data dictionaries delineate the fundamental entities required for system operation.

Entity: Tbl_Establishment_Profile

This primary table serves as the master registry for all commercial entities operating within the food supply chain, anchoring all subsequent risk calculations and inspection histories.

Attribute Name	Data Type	Constraint	Description
Establishment_ID	UUID	Primary Key	Cryptographically unique system identifier for the operator.
Local_Authority_Reg_No	Varchar(50)	Unique, Not Null	The official municipal trade license or registration number.
Legal_Entity_Name	Varchar(150)	Not Null	The formal registered trading name of the commercial enterprise.
FAO_Category_Code	Varchar(10)	Foreign Key	References the standardized categorization matrix (e.g., A2, H2, I1) dictating inherent hazard profiles.
Spatial_Coordinates	Geometry	Nullable	Longitude and latitude data enabling GIS mapping and spatial cluster analysis.
Certification_Type	Varchar(50)	Nullable	Logs validated quality systems (e.g., HACCP, ISO 22000, GMP) impacting inspection frequency.

Certification_Expiry	Date	Nullable	The explicit date upon which the system revokes the reduced inspection interval.
Dynamic_Risk_Index	Enum	Not Null	The continuously calculated status: High, Medium, Low, or Critical.
Jurisdictional_PHI_ID	UUID	Foreign Key	Assigns the establishment to the specific officer responsible for its oversight.

Entity: Tbl_H800_Inspection_Record

This table digitizes the structural elements of the legacy paper-based assessment, capturing granular data points necessary for automated compliance grading.

Attribute Name	Data Type	Constraint	Description and Regulatory Context
Inspection_Event_ID	UUID	Primary Key	Unique identifier for each specific site visit.
Establishment_ID	UUID	Foreign Key	Links the assessment data to the specific business profile.
Authorized_Officer_ID	UUID	Foreign Key	Identifies the specific inspector executing the assessment, ensuring accountability.
Execution_Timestamp	Datetime	Not Null	The exact date and time the assessment was initiated on.
Score_General_Env	Integer	Range(0-20)	Quantified assessment of overall premises hygiene and surrounding conditions.

Score_Infrastructure	Integer	Range(0-20)	Evaluation of structural soundness, drainage efficacy, and pest exclusion mechanisms.
Score_Personnel	Integer	Range(0-20)	Assessment of employee hygiene, uniform compliance, and verification of medical fitness.
Score_Processing	Integer	Range(0-20)	Evaluation of operational flow, cross-contamination barriers, and thermal processing logs.
Score_Storage_Trans	Integer	Range(0-20)	Assessment of cold chain integrity, raw/cooked segregation, and FIFO inventory rotation.
Cumulative_Percentage	Decimal(5,2)	Range(0-100)	The algorithmic sum of modular scores, dictating the ultimate regulatory standing.
Regulatory_Grade	Char(1)	Not Null	Automatically assigned outcome: A (75-100), B (50-74), C (25-49), D (0-24). ¹⁴
Inspector_Signature	Blob	Not Null	A cryptographic hash representing the officer's legal validation of the recorded facts.

Entity: Tbl_Sample_Custody_Register

Replicating and expanding upon the manual registers defined in the BESPA-FOOD guidelines, this table ensures the unassailable legal traceability of all physical evidence collected.

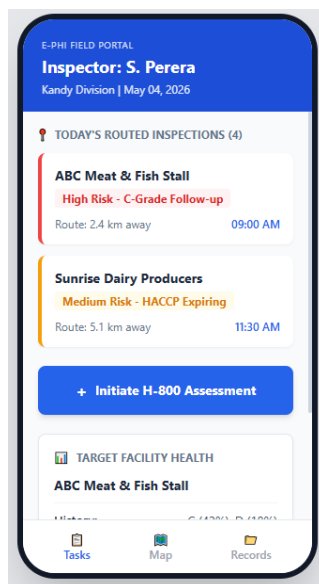
Attribute Name	Data Type	Constraint	Description and Regulatory Context
Sample_Record_ID	UUID	Primary Key	Internal system identifier for database normalization.
Sample_Reference_No	Varchar(50)	Unique, Not Null	Formatted, human-readable tracking alphanumeric (e.g., 2026/PHI/Matale/012).
Establishment_ID	UUID	Foreign Key	The exact origin point of the physical sample collection.
Commodity_Description	Varchar(150)	Not Null	Specific description of the item (e.g., UHT Pasteurized Milk).
Batch_Lot_Trace_No	Varchar(75)	Not Null	The manufacturer's production identifier, absolute requisite for targeted recalls.
Sampling_Objective	Enum	Not Null	Categorizes the action: Routine Monitoring, Targeted Surveillance, Compliance Enforcement.
Target_Laboratory_ID	UUID	Foreign Key	The accredited facility receiving the sample (e.g., Medical Research Institute).
Transfer_Timestamp	Datetime	Not Null	The moment physical custody is transferred from the field to the logistics/lab chain.

Analytical_Conclusion	Text	Nullable	Detailed scientific findings parsed from the laboratory API payload (e.g., "Salmonella detected in 25g").
Regulatory_Status	Enum	Nullable	Automatically flagged: Compliant, Non-compliant, Critical Hazard.
Enforcement_Action	Text	Nullable	The consequent regulatory response (e.g., Immediate Seizure, Court Prosecution, Written Warning).

11.4 Proposed Analytical Outputs and Management Dashboards

The immense volume of data generated by thousands of daily field inspections, laboratory results, and automated risk calculations is operationally useless without sophisticated, targeted visualization. The software module provides customized, role-based analytical outputs that translate raw data into actionable epidemiological intelligence.

11.4.1 Operational Field Topography Dashboards

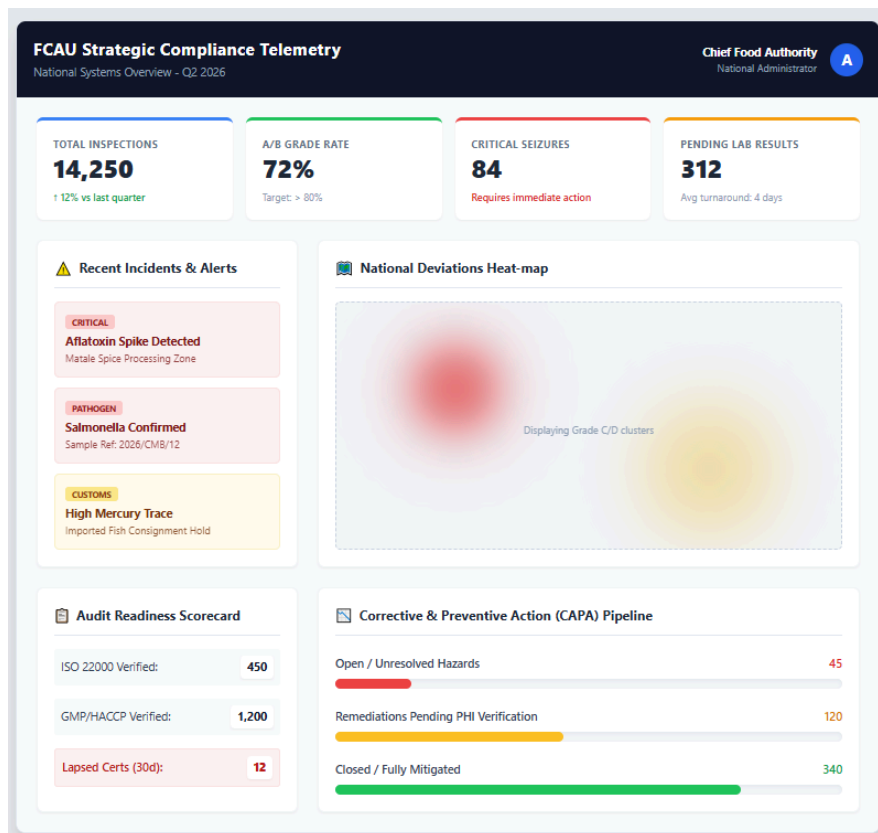


For the frontline workforce; Public Health Inspectors and local Medical Officers of Health, the system prioritizes tactical efficiency through the **Task Generation and Routing Dashboard**. This mobile optimized interface utilizes the dynamic scheduling engine to automatically generate daily itineraries, surfacing establishments whose inspection frequencies have lapsed or those flagged due to expiring certifications. The dashboard integrates overlaying the inspector's path with real time alerts if a nearby establishment suddenly escalates to a "Critical" status due to a positive lab result.

Crucially, the interface provides a "Facility Health Summary" screen. Prior to entering a premise, the inspector accesses a digitized brief detailing the establishment's previous five H-800 compliance grades, a chronological list of unresolved critical violations, and the status of any pending laboratory samples. This ensures the inspector enters the facility fully equipped

with historical context, allowing them to immediately target known vulnerabilities rather than conducting a generalized, unfocused audit.

11.4.2 Strategic Management and Compliance Telemetry



For regional directors, the Chief Food Authority, and national FCAU administrators, the software aggregates systemic data into the **Risk Management and Compliance Telemetry Dashboard**. This interface is designed to track macro-level Key Performance Indicators (KPIs) necessary for strategic resource allocation, policy formulation, and international reporting.

The primary output is a suite of interactive **Risk**

Heatmaps. By utilizing spatial characteristics and geocoordinates, the system renders a visual representation of the nation, highlighting dense clusters of Grade C and Grade D establishments.

Furthermore, the dashboard features an **Inspection Coverage and Efficiency Tracker**. This analytical tool automatically compares the algorithmic mandate for inspections (e.g., the system dictates that 1,200 high-risk inspections are required in the Western Province this quarter based on risk profiles) against the actual execution rate uploaded by the field inspectorate. This disparity analysis is critical for identifying workforce inefficiencies, justifying requests for increased departmental funding, or identifying regions requiring targeted capacity building.

12. Complaint Management Module

The Complaint Management Module enables the public to report suspected food safety or regulatory compliance violations through a structured, traceable, and accountable workflow. It functions as a **ticketing system** that captures complaints, assigns a unique reference number, routes cases to the appropriate jurisdiction (MOH/PHI), and tracks actions through to resolution.

Aligned with the system's end-to-end regulatory workflows (registration → inspection → enforcement), this module integrates with inspection, risk, and enforcement services to ensure that complaints trigger timely investigation and appropriate corrective or legal actions.

- Provide a simple, accessible channel for the public to lodge complaints
- Ensure each complaint is uniquely tracked with a reference number
- Route complaints automatically to the correct MOH based on GN division, FCAU to moderate it
- Enable MOH/PHI to investigate, act, and close cases
- Maintain full lifecycle visibility, auditability, and communication with complainants

12.1 Complaint Categories (Configurable)

- Food hygiene / sanitation issues
- Sale of expired or unsafe food
- Adulteration / contamination
- Improper storage / temperature violations
- Illegal or unregistered premises
- Labeling / packaging violations
- Consumer illness reports (foodborne)
- Others (custom category with description)
- Technical Matters

12.2 Functional Components

12.2.1 Public Complaint Intake

- Web form with:
 - Complainant details (optional anonymity supported)
 - Location (address + GN division or map pin)
 - Complaint category
 - Description
 - Attachments (images/documents)
- CAPTCHA / spam protection

12.2.2 Ticketing Engine

- Generates **unique reference number** (e.g., CMP-2026-000123)
- Creates a complaint record with initial status: OPEN
- Stores metadata for tracking and audit

12.2.3 Routing Engine (Geo + Rules)

- Uses **GN Mapping** to determine jurisdiction
- Assigns complaint to:
 - Relevant **MOH**
 - Optionally pre-assigns **PHI** (based on load/availability)
- Supports manual reassignment by FCAU

12.2.4 Case Management (MOH/PHI)

- MOH dashboard:
 - View assigned complaints
 - Assign to PHI

- Set priority (Low/Medium/High/Critical)
- PHI actions:
 - Schedule inspection
 - Record findings
 - Attach evidence
 - Recommend actions

12.2.5 Investigation & Enforcement Integration

- Trigger inspection workflow
- Link complaint to:
 - **Inspection records (H800)**
 - **Laboratory sampling** (if needed)
 - **Enforcement actions** (warnings, penalties, prosecution)
- Outcome updates reflected back in the complaint ticket

12.2.6 Status Tracking & Communication

- Status transitions:
 - OPEN → ASSIGNED → IN_PROGRESS → RESOLVED → CLOSED
 - Optional: REJECTED, DUPLICATE
- Notifications:
 - Reference number issued (on submission)
 - Status updates
 - Resolution summary

13. Notifications Module

The Notifications Module is a cross-cutting service within the IIMS responsible for delivering timely, reliable, and context aware communications to all stakeholders, including Food Business Operators (FBOs), Public Health Inspectors (PHIs), Medical Officers of Health (MOHs), laboratory personnel, and administrative authorities. The module ensures that critical system events such as application submissions, inspection outcomes, appeal decisions, and compliance actions are communicated effectively through **in-system notifications** and **email alerts via SMTP**.

Aligned with the workflow driven nature of the system described in the SRS, this module plays a vital role in maintaining transparency, operational continuity, and user engagement by ensuring that all actors are informed of relevant updates in real time or near real time.

- Provide real-time and asynchronous notifications for key system events
- Ensure reliable email delivery using SMTP infrastructure
- Support role-based and event-driven notification targeting
- Enhance user awareness, responsiveness, and compliance
- Maintain a complete audit trail of all notifications sent

13.1 Notification Types

System Notifications (In-App)

- Displayed within the application dashboard
- Accessible via notification panel
- Persisted in database for tracking and audit

Email Notifications (SMTP-Based)

- Delivered to registered email addresses
- Used for critical or external communication
- Supports formatted templates (HTML)

13.2 Functional Components

Notification Engine

- Central service responsible for:
 - Event listening (e.g., APPLICATION_SUBMITTED, INSPECTION_COMPLETED)
 - Triggering notification workflows
- Implements event-driven architecture using queues

Template Management

- Predefined templates for different notification types
- Supports:
 - Dynamic placeholders (e.g., {user_name}, {application_id})

Delivery Channels

- **In-App Channel**
 - Stored in notifications table
 - Mark as read/unread
- **Email Channel (SMTP)**
 - Uses Laravel Mail system
 - Supports SMTP servers (e.g., Gmail, SendGrid, Gov SMTP)

User Preference Management

- Users can configure:
 - Preferred notification channels
 - Opt-in/out for non-critical alerts

Notification Logging & Audit

- All notifications stored with:
 - Timestamp
 - Delivery status (sent, failed, pending)
 - Recipient details
- Supports compliance and troubleshooting

14. Reports Module

The Reports Module is a strategic component of the IIMS that transforms operational data into meaningful insights for monitoring, decision-making, and regulatory governance. It consolidates data generated across all system modules such as premises registration, inspections, laboratory results, enforcement actions, and risk assessments into structured reports and analytical dashboards.

Given the system's mandate to support national-level food safety oversight, the Reports Module enables stakeholders at different levels (FCAU administrators, MOHs, PHIs, and policymakers) to evaluate compliance trends, identify risks, and take informed actions based on real-time and historical data.

- Provide real-time and historical reporting capabilities
- Support data-driven decision-making at operational and strategic levels
- Enable monitoring of compliance, risk, and enforcement activities
- Facilitate regulatory transparency and accountability
- Support export and sharing of reports for audits and governance

14.1 Report Categories

Operational Reports

- Daily/weekly inspection reports
- Pending applications and approvals
- Inspection schedules and completion status

Compliance Reports

- Grading distribution (A, B, C, D)
- Non-compliance trends
- Corrective actions and follow-ups

Risk-Based Reports

- High-risk premises identification
- Risk categorization trends
- Inspection frequency analysis

Laboratory Reports

- Sample testing results
- Hazard detection trends
- Lab processing timelines

Enforcement Reports

- Violations and penalties
- Prosecution records
- License suspensions and closures

Import/Export Reports

- Import inspections and rejections
- Export certifications issued
- Border compliance statistics

14.2 Functional Components

Reporting Engine

- Aggregates data from multiple services
- Supports both real-time queries and pre-aggregated data
- Enables filtering, grouping, and sorting

Dashboard Interface

- Role-based dashboards:
 - **FCAU** → National-level analytics
 - **MOH** → Regional compliance insights
 - **PHI** → Field-level performance
- Visualizations:
 - Charts (bar, line, pie)
 - Heatmaps (GN-based risk distribution)
 - KPIs (inspection counts, compliance rates)

Report Builder

- Allows users to:
 - Select report type
 - Apply filters (date range, location, category)
 - Customize output

Export & Sharing

- Formats:
 - PDF
 - Excel (XLSX)
 - CSV
- Scheduled report generation (e.g., monthly reports emailed)

15. Minimum Viable Product (MVP) Triage and Scope Definition

An effective Minimum Viable Product must isolate the absolute core functionality that delivers immediate operational value while serving as the stable foundation for subsequent iterative expansions.

To determine the MVP scope, the modules were evaluated against three criteria: statutory criticality, inter-agency dependency risk, and architectural foundational necessity.

Module Name	Architectural Justification
User and Access Management	Absolute prerequisite. No other module can function without authenticated identities, RBAC, and secure session orchestration.
Premises Registration	Foundational data layer. The system must establish a registry of physical entities and spatial routing logic before any field operations can occur.
Inspection and Grading	The core operational mandate of the FCAU. Digitizing the H-800 form provides immediate, visible efficiency gains for the field inspectorate.
Laboratory Coordination	Involves complex API integrations with varied external LIMS systems and unreliable OCR parsing for legacy PDFs. Too high a risk module with technical dependencies.
Legal and Enforcement	Requires integration with judicial workflows. Temporary enforcement actions can be managed administratively until the core inspection engine is stabilized.
Import / Export Control	Mandates high-risk, cross-agency API synchronization with the National Single Window, ASYCUDA World, and digital payment gateways (IPG). Requires extensive external testing.
Salt / Bottled Water Mgmt.	Highly specialized parent-child architectures and dynamic distance-based fee structures introduce the complexity during the initial build.
Risk Analysis Framework	Predictive algorithmic modulation requires a vast repository of historical inspection and laboratory data to function accurately. The MVP must first generate this data corpus.

16. Non-Functional Requirements

While functional requirements dictate the specific workflows the IIMS must perform, the Non-Functional Requirements (NFRs) define the systemic constraints, quality attributes, and architectural parameters necessary to ensure the platform operates securely, reliably, and efficiently at a national scale. Aligning these NFRs is critical for a governmental platform subject to public scrutiny and legal compliance.

16.1 Security and Identity Protection

As the repository for national commercial data, geospatial intelligence, and legally sensitive enforcement records, the IIMS must adhere to the highest standards of cybersecurity. The architecture must assume a hostile operating environment and implement defense-in-depth strategies.

Security NFR	Technical Specification	Operational Impact
Data Confidentiality	All sensitive data in transit must be encrypted utilizing TLS 1.3 protocols. Data at rest, specifically user credentials, NIC numbers, and proprietary compliance documents, must be encrypted using AES-256 or similar standards within the shared MySQL database.	Protects citizen and commercial data from exposure in the event of an infrastructure breach or interception.
Authentication Strength	Mandatory implementation of Multi-Factor Authentication (MFA) combining knowledge factors (passwords) with possession factors (Time-based One-Time Passwords or cryptographic email links).	Drastically reduces the attack surface associated with credential stuffing, phishing, or brute-force attacks against health officers.
Access Control	Strict enforcement of Role-Based Access Control (RBAC) ensuring users are granted the absolute minimum privileges required to execute their statutory duties.	Prevents a compromised FBO account from accessing administrative dashboards or altering inspection data.

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Vulnerability	Description	Preventive Actions
Broken Access Control	How a web application grants access to content and functions to some users and not others	Access control user management
Cryptographic Failures	Cryptographic failures occur when an application does not adequately protect sensitive information which often lead to exposure of sensitive data	Data encryption
Injection	Injection can result in data loss, corruption, or disclosure to unauthorized parties, loss of accountability, or denial of access	Keeping data separate from commands and queries Use safe API Server-side input validation
Insecure Design	Insecure design focuses on risks related to design and architectural flaws, with a call for more use of threat modeling, secure design patterns, and reference architectures	Establish and use a library of secure design patterns or paved road ready to use components Use threat modeling for critical authentication, access control, business logic, and key flows Integrate security language and controls into user stories
Security Misconfiguration	Typically occur when holes are left in the security	Identically configure development, staging and

	framework of an application by systems administrators, DBAs or developers	production environments with different credentials
Vulnerable and Outdated Components	Vulnerable Components are a known issue that we struggle to test and assess risk and is the only category to not have any Common Vulnerability and Exposures	Remove unused dependencies, unnecessary features, components, files, and documentation. Only obtain components from official sources over secure links.
Identification and Authentication Failures	Confirmation of the user's identity, authentication, and session management is critical to protect against authentication related attacks	Where possible, implement multifactor authentication to prevent automated credential stuffing, brute force, and stolen credential reuse attacks. Do not ship or deploy with any default credentials, particularly for admin users. Implement weak password checks, such as testing new or changed passwords.
Software and Data Integrity Failures	Software and data integrity failures relate to code and infrastructure that does not	Use e-signatures or similar mechanisms to verify the software or data is from the

	protect against integrity violations	expected source and has not been altered.
Security Logging and Monitoring Failures	Detect, escalate, and respond to active breaches. Without logging and monitoring, breaches cannot be detected. Insufficient logging, detection, monitoring, and active response occurs any time	Ensure all login, access control, and server-side input validation failures can be logged with sufficient user context to identify suspicious or malicious accounts and held for enough time to allow delayed forensic analysis Ensure that logs are generated in a format that log management solutions can easily consume
Server-Side Request Forgery	SSRF flaws occur whenever a web application is fetching a remote resource without validating the user-supplied URL	Sanitize and validate all client supplied input data Enforce the URL schema, port, and destination with a positive allow list

16.2 Performance

The performance of the Interoperable Risk-Based Food Safety Inspection Information Management System (IIMS) is a critical quality attribute that directly influences its effectiveness in supporting nationwide regulatory operations. Given the system's role in managing high volume, multi actor workflows; including premises registration, inspection scheduling, real time field data capture, laboratory integrations, and enforcement actions. It must consistently deliver fast, reliable, and scalable responses under varying load conditions.

The system is expected to support concurrent access by diverse user groups such as Food Business Operators (FBOs), Public Health Inspectors (PHIs), Medical Officers of Health (MOHs), laboratory personnel, and administrative authorities, often operating simultaneously across different geographical regions. Additionally, several core processes such as automated risk categorization, GN division mapping, duplicate validation, and inspection grading are computationally intensive and must be executed efficiently without degrading overall system responsiveness.

To ensure seamless user experience and operational continuity, the system must adopt a performance-oriented architecture that leverages asynchronous processing, optimized database interactions, caching strategies, and scalable infrastructure. Special consideration must also be given to mobile responsive field operations, where PHIs may operate under constrained network conditions and require near real time system feedback.

Overall, performance requirements are defined to guarantee that the system can handle peak workloads, maintain low latency for critical transactions, and support future scalability as the platform expands across additional regulatory domains and integrations.

The bandwidth is assumed at 1mbps (shared) with 1,000 concurrent users (50% load factor) in total.

Action	Performance Criteria
Screen Navigation: field-to-field	< 5 milliseconds
Screen Navigation: screen-to-screen	< 3 seconds
Screen Refresh	< 3 seconds
Screen list box, combo box	< 2 seconds
Screen grid – 25 rows, 10 columns	<3 seconds
Report preview– (all reports) – initial page view (if asynchronous)	< 40 seconds in most instances. For complicated / large volume reports may require a longer period
Simple query single table, 5 fields, 3 conditions – without screen rendering	< 4 seconds for 10,000 rows
Complex query – multiple joined table (5), 10 fields, 3 conditions – without screen rendering	< 6 seconds for 10,000 rows
Server-side validations / computations	< 10 milliseconds
Client-side validations / computation	< 5 milliseconds
Batch processing (if any) per 100 records	< 120 seconds
Login, authentication, and verification	< 3 seconds

Data Archive

As a part of the system performance criteria, historical data archive is a mandatory requirement. Automated data archive will be handled by the system as below.

- Inactive user profiles which are not accessed the system for 01 year
- Notifications log which are later than 03 months
- User login log which are later than 03 months

16.3 Scalability and Availability

The IIMS must remain highly responsive and available to support continuous field operations by PHIs across diverse geographical regions, including areas with suboptimal network connectivity.

Technical Specification		Operational Impact
System Availability	The infrastructure must be designed to achieve a 99.9% uptime Service Level Agreement (SLA). This requires redundant application deployment and automated database failover mechanisms.	The system must be highly available to ensure uninterrupted execution of critical regulatory functions. Continuous availability allows inspectors to conduct and record inspections in real time, enables authorities to review and approve decisions without delay, and allows businesses to complete registrations and compliance activities on time. Any downtime can disrupt inspection workflows, delay approvals, and weaken risk based monitoring, ultimately affecting public health oversight. Therefore, high availability is essential to maintain operational continuity, regulatory efficiency, and trust in the system.

Granular Scalability	The microservices architecture must allow independent horizontal scaling. For example, the Inspection Service must scale autonomously during periods of intense nationwide compliance audits without requiring identical resource allocation.	Optimizes cloud computing expenditure by dynamically allocating CPU and memory resources exclusively to the subsystems experiencing acute load.
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16.4 Maintainability and Auditability

For the FCAU to effectively govern the platform and continuously improve its regulatory strategies, the system architecture must be highly observable and historically immutable.

	Technical Specification	Operational Impact
Traceability	Implementation of centralized, distributed tracing utilizing correlation IDs injected at the API Gateway, tracking requests as they traverse multiple microservices.	Allows technical administrators to rapidly isolate bottlenecks or failures in cross-service communications.
Accountability Logging	Comprehensive logging of all critical state-machine transitions and database mutations, recording the actor, timestamp, original value, and modified value.	Establishes an unassailable forensic trail to investigate administrative errors or suspected corruption in the grading process.
Concurrency Management	Utilizing ACID-compliant database transactions to manage simultaneous data interactions.	Prevents data corruption when multiple actors (e.g., an FBO uploading an amendment while an MOH reviews the profile) access the same record.

16.5 Interoperability

Interoperability is a foundational requirement of the IIMS, as the system is designed to operate within a multi-agency, data-driven regulatory ecosystem. The platform must seamlessly exchange data with internal modules as well as external stakeholders such as laboratories, import/export authorities, and other government systems. Given that the SRS explicitly emphasizes an *interoperable, integrated system* supporting end-to-end food safety governance, interoperability ensures that information flows accurately, securely, and in near real time across all participating entities.

Effective interoperability reduces duplication of data entry, minimizes manual intervention, and enables coordinated decision-making across organizational boundaries. It is especially critical for workflows such as laboratory result ingestion, GN-based routing, risk-based inspection scheduling, and enforcement actions, where multiple systems and actors must interact seamlessly.

Interoperability Criteria	Technical Specification	Operational Impact
Standard Data Exchange Formats	Use JSON (primary) and XML (optional for legacy systems); UTF-8 encoding	Ensures consistent data interpretation across systems and avoids integration errors
API-Based Integration	RESTful APIs with versioning (/api/v1/), OpenAPI/Swagger documentation	Enables seamless integration with external systems such as regulatory bodies
Authentication & Secure Integration	OAuth2 / JWT-based authentication, API gateway, HTTPS	Protects sensitive regulatory and personal data during inter-system communication
Asynchronous Communication	Message queues for lab results, notifications, background engines	Prevents system delays and supports high-volume data exchange without blocking operations

Geospatial Interoperability	Integration with GIS services for GN division mapping and spatial routing	Ensures accurate assignment of inspections and approvals
Data Consistency & Validation	Central validation services (duplicate detection, schema validation)	Prevents conflicting records and maintains integrity across multiple systems
File & Document Exchange	Secure file storage (e.g., object storage), API-based upload/download, malware scanning	Enables safe sharing of compliance documents and laboratory reports
Event-Driven Integration	Publish/subscribe model for events (e.g., application submitted, inspection completed)	Supports real-time updates and automated workflow triggering across modules
Backward Compatibility	API versioning and deprecation strategy	Allows external systems to evolve without breaking integrations
Monitoring & Logging	API logs, integration monitoring dashboards, error tracking	Facilitates troubleshooting and ensures reliability of integrations

16.6 Usability

Usability is a critical quality attribute of the IIMS, as the system is intended to be used by a diverse group of stakeholders with varying levels of technical expertise, including Food Business Operators (FBOs), Public Health Inspectors (PHIs), Medical Officers of Health (MOHs), laboratory personnel, and administrators. The system must provide an intuitive, efficient, and accessible user experience that enables users to complete their tasks with minimal effort and error.

Given the operational context defined in the SRS such as field inspections, progressive onboarding, multi-step workflows, and document handling, the system must prioritize

simplicity, clarity, and responsiveness. Special attention is required for mobile-responsive interfaces used by PHIs during on-site inspections, as well as guided workflows for FBOs who may have limited digital literacy.

Effective usability reduces training requirements, minimizes user errors, accelerates task completion, and increases adoption and trust in the system. It also ensures that regulatory processes are executed accurately and efficiently without unnecessary friction.

Usability Criteria	Technical Specification	Operational Impact
Intuitive User Interface	Clean UI with logical navigation, consistent layouts, and clear labels (Laravel Blade templates)	Reduces learning curve and enables users to perform tasks quickly and accurately
Role-Based Dashboards	Dynamic dashboards tailored to roles (FBO, PHI, MOH, Admin)	Improves efficiency by showing only relevant information and actions per user
Mobile Responsiveness	Responsive design (CSS frameworks like Bootstrap/Tailwind) for mobile and tablet devices	Enables PHIs to conduct inspections and submit data in the field without usability issues
Progressive Onboarding	Multi-step registration forms with validation and save-as-draft capability	Reduces user abandonment and improves completion rates for applications
Form Validation & Error Handling	Real-time validation, clear error messages, and guided corrections	Minimizes data entry errors and ensures data quality
Accessibility	Basic WCAG compliance (readable fonts, contrast, keyboard navigation)	Ensures usability for a wider range of users, including those with accessibility needs

Multilingual Support	Support for Sinhala, Tamil, and English for Public UIs	Enhances inclusivity and usability across diverse user groups in Sri Lanka
Performance Feedback	Loading indicators, success confirmations, and status tracking	Keeps users informed and improves confidence in system operations

17. Constraints and Assumptions

17.1 System Constraints

System constraints define the limitations and conditions under which the system must be designed and implemented.

Technical Constraints

Technology Stack Constraints

- Backend will be implemented using Laravel (PHP framework)
- No frontend framework (e.g., React, Angular) is to be used; UI will rely on:
 - Laravel Blade templates
 - Bootstrap Library
 - Server-side rendering
- Database will be MySQL (relational model)
- Architecture will follow microservices-based design with a shared database

Architecture Constraints

- Microservices will be logically separated by domain (e.g., Inspection, Complaints, Reports)
- Despite microservices, a **shared database schema** is enforced

- Services will communicate via:
 - REST APIs
 - Asynchronous queues (for events such as notifications, lab results)
- Centralized authentication will be used across all services

Integration Constraints

- System will integrate with:
 - Laboratory systems (API + PDF ingestion)
 - Email systems via **SMTP**
 - GIS/GN division mapping services
- All integrations will use **secure REST APIs (HTTPS)**
- External systems may have limited standardization → require adaptable API design

Data Constraints

- Use a **relational schema** aligned with defined entities (e.g., applications, inspections, complaints)
- Strong data integrity will follow (foreign keys, validation rules)
- File storage will support:
 - Large document uploads
 - Malware scanning
- Audit logs will be persistently stored

Data Sharing & Interoperability Constraints

- Support interoperability with:
 - External agencies
 - Internal modules
- Data exchange formats limited to:

- JSON (primary)
 - XML (optional)
- API versioning required to maintain backward compatibility

Operational Constraints

User Constraints

- Users include non-technical stakeholders (FBOs, field inspectors)
- Interfaces will be:
 - Simple
 - Mobile-responsive

17.2 System Assumptions

Assumptions define expected conditions that are considered true for system design and operation.

Technical Assumptions

- Laravel framework is sufficient to support microservices architecture
- MySQL database can handle:
 - Transactional workloads
 - Moderate analytical queries
- Queue system will be available for async processing
- SMTP services will be reliable for email delivery

Integration Assumptions

- External laboratory systems will:

- Provide APIs or standardized reports
- GIS/GN mapping data will be:
 - Accurate and maintained
- External agencies will support secure API communication

Data Assumptions

- Data provided by users (FBOs, public complaints) is:
 - Generally accurate but requires validation
- Required regulatory data structures (H800, categories) are stable

User Assumptions

- Users will have:
 - Basic digital literacy
- PHIs will use:
 - Mobile devices during inspections
- Users will follow defined workflows and procedures

Operational Assumptions

- FCAU will manage:
 - System configurations (risk rules, grading criteria)
- MOH and PHI will:
 - Act on assigned tasks within defined timelines
- Complaint handling will follow:
 - Official escalation and enforcement processes

Infrastructure Assumptions

- Hosting environment will provide:
 - Reliable uptime
 - Backup and recovery mechanisms
- Internet connectivity will be available for all operations
- System will be deployed in a scalable environment (cloud or equivalent)

Security Assumptions

- Users will maintain confidentiality of credentials
- External integrations will follow secure communication protocols
- No malicious insider threats beyond standard risk assumptions

Reporting Assumptions

- Stakeholders will use reports for:
 - Monitoring and decision-making
- Data volume will grow over time, requiring optimization

17.3 Scope Constraints

- For all above external API integrations mentioned under the functional requirements list, client should facilitate with relevant API documentations, and if any APIs are not available or not feasible, Human in the loop walk around will facilitate within the system.
- GIS map integration can be considered through Google Map API if the operational and contractual obligations are sorted out.
- All publicly available UIs will be supported with 03 Languages (EN, SIN, TAMIL) as the visual labels, but all data entry would be handled only through EN.
- Report generation would be based on static templates where dynamic configuration will not be part of it, but advanced filtering options will be accommodated.
- There will be no integration with third party email services where number of emails per day would be a constraint.

- System will fully operate under the stable Internet connection only where offline synchronization would not be a part of here.
- Deployment infrastructure should be mapped as per the FRs and NFRs discussed here.
- Development of LIMS is not a part of the scope.

18. Approval and Sign-off of the Software Requirement Specification (SRS)

Sign-off the above described software Requirement Specification (SRS)

.....

Authorized Representative – Food Control Administrative Unit (FCAU)

Name :

Designation :

Date :

.....

Authorized Representative – United Nation Food and Agriculture Organization (UNFAO)

Name :

Designation :

Date :

.....

Authorized Representative – Procons Infotech (Pvt) Ltd

Name :

Designation :

Date :

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